



Monetary Authority
of Singapore

FINANCIAL STABILITY REVIEW



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Economic Policy Group
November 2025

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Macroprudential Surveillance Department
Economic Policy Group
Monetary Authority of Singapore
10 Shenton Way
MAS Building
Singapore 079117

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Definitions and Conventions

As used in this report, the term “country” does not in all cases refer to a territorial entity that is a state as understood by international law and practice. As used here, the term also covers some territorial entities that are not states but for which statistical data are maintained on a separate and independent basis.

In this report, the following groupings are used:

- “ASEAN-5” comprises Indonesia, Malaysia, Philippines, Singapore and Thailand.
- “AE” refers to advanced economies comprising Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Ireland, Israel, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom and United States.
- “Asia-9” refers to China, India, Indonesia, Malaysia, Philippines, South Korea, Taiwan, Thailand and Vietnam. “Asia-8” refers to the same countries, excluding China.
- “EM Asia” comprises Asian economies such as China, Hong Kong, India, Indonesia, Malaysia, Singapore, South Korea and Thailand.
- “EM” refers to emerging economies comprising EM Asia economies as well as Argentina, Brazil, Chile, Colombia, Czech Republic, Hungary, Mexico, Poland, Russia, Saudi Arabia, South Africa and Türkiye.
- “Eurozone” comprises Austria, Belgium, Cyprus, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Portugal, Slovakia, Slovenia and Spain.
- “G3” comprises the Eurozone, Japan, and the United States.

Abbreviations used for financial data are as follows:

- Currencies: Australian Dollar (AUD), British Pound (GBP), Chinese Yuan (CNY), Euro (EUR), Hong Kong Dollar (HKD), Indonesian Rupiah (IDR), Indian Rupee (INR), Japanese Yen (JPY), Korean Won (KRW), Malaysian Ringgit (MYR), New Taiwan Dollar (TWD), New Zealand Dollar (NZD), Singapore Dollar (SGD), Thai Baht (THB), US Dollar (USD), Vietnamese Dong (VND).

Other abbreviations:

AE	Advanced Economy
AFS	Accommodations and Food Services
AI	Artificial Intelligence
AUM	Assets under Management
bps	Basis points
CAR	Capital Adequacy Ratio
CCB	Capital Conservation Buffer
CCP	Central Counterparty
CCyB	Countercyclical Capital Buffer
CEI	Composite Economic Indices
CET1	Common Equity Tier 1
CIS	Collective Investment Schemes
CIS Code	Code on Collective Investment Schemes
CISS	Composite Indicator of Systemic Stress
CO	Commodity

COVID-19	Coronavirus Disease 2019
CR	Credit
CRE	Commercial Real Estate
DDRS	Depository Trust & Clearing Corporation Data Repository (Singapore) Pte Ltd
DOS	Department of Statistics
D-SIBs	Domestic Systemically Important Banks
EC	Executive Condominiums
EM	Emerging Markets
EPG	Economic Policy Group
EPI	Export Price Index
EQ	Equity
ESG	Enterprise Singapore
ETI	Expenditure-to-Income
FCI	Financial Conditions Index
FI	Financial Institution
FSB	Financial Stability Board
FSI	Financial Stress Index
FSR	Financial Stability Review
FVI	Financial Vulnerabilities Index
FX	Foreign Exchange
GDP	Gross Domestic Product
GLS	Government Land Sales
HHI	Herfindahl-Hirschman Index
ICCA	Initial Cross-country Average
ICR	Interest Coverage Ratio
IR	Interest Rate
IWST	Industry-wide Stress Test
LCR	Liquidity Coverage Ratio
LLM	Large Language Models
LTD	Loan-to-deposit
MAS	Monetary Authority of Singapore
m-o-m	Month-on-month
MSD	Macroprudential Surveillance Department
NAV	Net Asset Value
NBFI	Non-bank Financial Institution
NFC	Non-financial Corporate
NPA	Non-performing Assets
NPISH	Non-Profit Institutions Serving Households
NPL	Non-performing Loan
NSFR	Net Stable Funding Ratio
NUS-CRI	National University of Singapore Credit Research Initiative
OIF	Offshore Insurance Fund
OTC	Over-the-Counter
PD	Probability of Default
PDI	Personal Disposable Income
PPI	Producer Price Index

PSTASSA	Professional, Scientific, Technical, Administrative, Support Service Activities
q-o-q	Quarter-on-quarter
RCR	Redemption Coverage Ratio
RHS	Right Hand Side
RWA	Risk-weighted assets
SERT	Singapore Economic Resilience Taskforce
SFA	Securities and Futures Act
SGX	Singapore Exchange
SIF	Singapore Insurance Fund
SME	Small and Medium-sized Enterprise
SORA	Singapore Overnight Rate Average
STI	Straits Times Index
TNA	Total Net Assets
URA	Urban Redevelopment Authority
VAR	Vector Autoregression
y-o-y	Year-on-year

Preface

The Monetary Authority of Singapore (MAS) conducts regular assessments of Singapore's financial system to identify potential vulnerabilities and review its resilience to potential shocks and risks. The analyses and results are published in the annual Financial Stability Review (FSR), which aims to contribute to a better understanding of issues affecting Singapore's financial system among market participants, analysts and the public.

Section 1 of the FSR provides a discussion of the risks in the external environment, while Section 2 assesses the financial conditions in Singapore. This is followed by an analysis of the Singapore corporate and household sectors in Sections 3 and 4. A review of the banking and non-bank financial institution (NBFI) sectors is then provided in Sections 5 and 6. This FSR, unless otherwise stated, reflects information available as of 27 October 2025.

The production of the FSR is coordinated and prepared by the Macroprudential Surveillance Department (MSD) of the Economic Policy Group (EPG) that comprises Ang Wei Han, Adrian Chiu, Cheryl Ho, Jerald Ho, Huang Zhengjie, Sherwin Lau, Wendy Lee, Leou Jie Dong, Aloysius Lim, Kalena Lim, Janice Loh, Low Yen Ling, Lye Royii, Ng Heng Tiong, Desmond Ong, Alex Phua, Edward Robinson, Celine Sia, Glenda Soh, Suah Jing Lian, Tan Shu Yan, Tan Wei Ting, Teoh Shi-Ying, Christopher Walker, Xiong Wei, Yeo Han Sia and Davis Yep.

The FSR also includes contributions from other MAS departments including Banking Departments I, II & III, Corporate Finance & Disclosures Department, Economic Analysis Department, Economic Surveillance & Forecasting Department, Enterprise Knowledge Department, Insurance Department, Investment Intermediaries Department, and Prudential Policy Department.

The FSR may be accessed in PDF format on the MAS website:

<https://www.mas.gov.sg/publications/financial-stability-review>

Overview

Financial stability risks remain elevated despite economic resilience

The global economy has shown greater than expected resilience this year despite a challenging trade and geopolitical backdrop. Global growth has been supported by the front-loading of demand, strong technology-related investments, and accommodative financial conditions. Amid moderating inflation, many central banks—notably in emerging markets (EMs)—have reduced policy rates, as broad-based easing trends in the dollar provided additional monetary space to support growth.

Globally, financial stability risks are elevated despite the buoyant sentiments across several financial markets. Some equity markets are seeing relatively stretched valuations, particularly in the technology and artificial intelligence (AI) segments. In the sovereign bond markets, there are concerns over fiscal sustainability amid high and rising public debt levels. Recent prominent credit losses involving private credit funds have drawn attention to incipient signs of rising corporate credit risks, which could trigger a reversal of historically low corporate bond spreads.

Against slower growth and a possible resurgence in local currency weakness and capital outflows, regional central banks will face complex trade-offs between price and external stability. The relative resilience demonstrated by EMs during recent global financial cycle fluctuations highlighted the critical role played by credible macroeconomic policy responses and sufficient foreign exchange (FX) reserves. Policymakers must therefore remain alert to potential disruptions whilst ensuring adequate policy flexibility to respond swiftly when needed.

Strong financial positions have supported the resilience of corporates, households and the financial sector in Singapore

Corporates, households and banks in Singapore have remained resilient over the past year. Singapore's financial conditions have been mildly supportive with interest rates and hence borrowing costs declining amid expansions in credit and money supply. Overall bank credit quality was strong as most corporates benefitted from the firmer-than-expected domestic growth momentum, while households were supported by stable income growth.

- The financial positions of corporates in Singapore have strengthened over the past year despite a surge in uncertainty and market volatility in April, which has since receded. Firms' debt-servicing capacity has improved alongside resilient corporate earnings and a more accommodative financing environment. Strong earnings have also enabled firms to continue to accumulate cash buffers.
- Households have remained financially resilient over the past year, supported by firm financial asset growth. Debt-servicing capacity has been healthy, underpinned by stable income growth and lower mortgage rates.

- The banking sector has maintained strong capital and liquidity positions. For the non-bank sector, insurers remained well-capitalised and investment funds managed liquidity risks well.

The economy's overall resident credit-to-GDP ratio declined from 143% in Q2 2024 to 137% in Q2 2025, driven by nominal GDP growth. Accordingly, the resident credit-to-GDP gap remained negative in Q2 2025. MAS will maintain the Countercyclical Capital Buffer (CCyB) at 0% for 2026.

Singapore corporates, households, and financial institutions should remain vigilant amid uncertainties in the external environment

MAS' stress tests show that the corporate and household sectors have adequate buffers to manage shocks to incomes and financing costs. Most firms and households are expected to stay resilient under stress given the current strength of their balance sheets. However, a small proportion of corporates and households that are more leveraged or have thinner buffers may face debt-servicing pressures and cashflow constraints. In view of the uncertain macroeconomic environment, businesses and households should remain vigilant to downside risks.

The results of the Industry-wide Stress Test (IWST) 2025 exercise affirmed that banks and insurers are in a strong position to withstand severe macrofinancial shocks. Nevertheless, banks should continue to maintain sound credit risk management practices and adequate provisioning buffers amid the uncertain macroeconomic environment. Risks may manifest as sharp repricing shocks, with the potential to cause spikes in investment fund redemptions or impose capital losses on insurers. Investment funds should thus continue to monitor liquidity risks while insurers should remain vigilant and closely monitor their capital and liquidity positions, and business strategy risks amid continuing market volatilities.

Macroprudential Surveillance Department, Economic Policy Group
Monetary Authority of Singapore
5 November 2025

1 Global Macrofinancial Environment

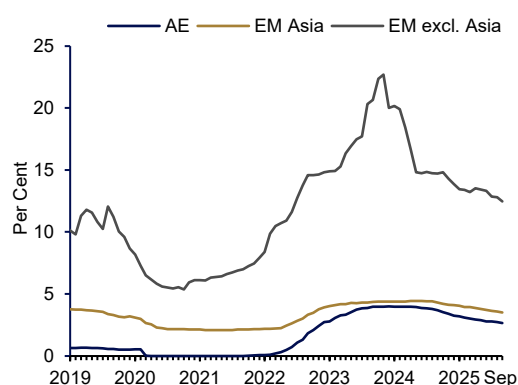
Elevated financial stability risks despite economic resilience

The global economy this year has shown more resilience than expected despite a challenging trade and geopolitical backdrop. Global economic activity has been supported by the front-loading of demand, strong technology-related investments and accommodative financial conditions. Amid moderating inflation, many central banks, notably EMs, have reduced policy rates, as broad-based trends in the dollar provided additional monetary space to support growth (**Chart 1.1**). However, risks to the outlook remain tilted to the downside. While measures of geopolitical risk and trade policy uncertainty have improved, they remain elevated relative to a year ago (**Chart 1.2**). Extreme tariffs and all-out retaliatory trade wars have been avoided thus far, but the potential for further volatility in trade restrictions could prolong firms' caution in making new investments outside the technology sector.

Financial stability risks are elevated despite prevailing favourable investment sentiments across several financial markets globally. Equity markets are seeing relatively stretched valuations concentrated in the technology and AI sectors. In sovereign bond markets, there are mounting concerns over fiscal sustainability amid high and rising public debt levels. Recent prominent credit losses involving private credit funds have drawn attention to incipient signs of rising corporate credit risks, which may trigger a reversal of historically low corporate bond spreads.

Chart 1.1 Central banks have reduced policy rates

Central bank policy rates

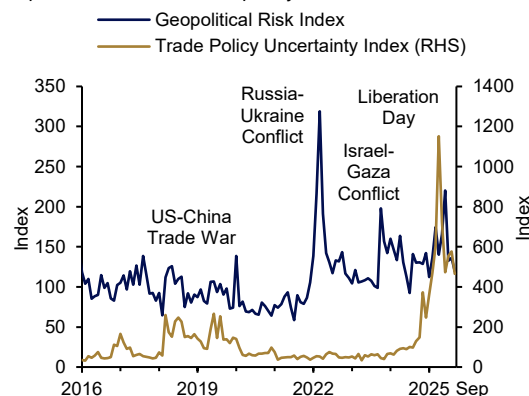


Source: Haver Analytics, MAS estimates

Note: Simple averages of central bank policy rates in each region. EM Asia excludes Singapore.

Chart 1.2 Geopolitical risk and trade policy uncertainty remain elevated

Geopolitical risk and trade policy uncertain indices¹



Source: Caldara and Iacoviello (2022); and Caldara, Iacoviello, Molliogo, Prestipino and Raffo (2020)

¹ Caldara, Dario, and Matteo Iacoviello. 2022. "Measuring Geopolitical Risk." *American Economic Review*, 112 (4): 1194–1225; Caldara, Dario, Matteo Iacoviello, Patrick Molliogo, Andrea Prestipino, and Andrea Raffo. 2020. "The Economic Effects of Trade Policy Uncertainty." *Journal of Monetary Economics*, 109: 38–59

Public equity markets remain euphoric, with growing concentration risk

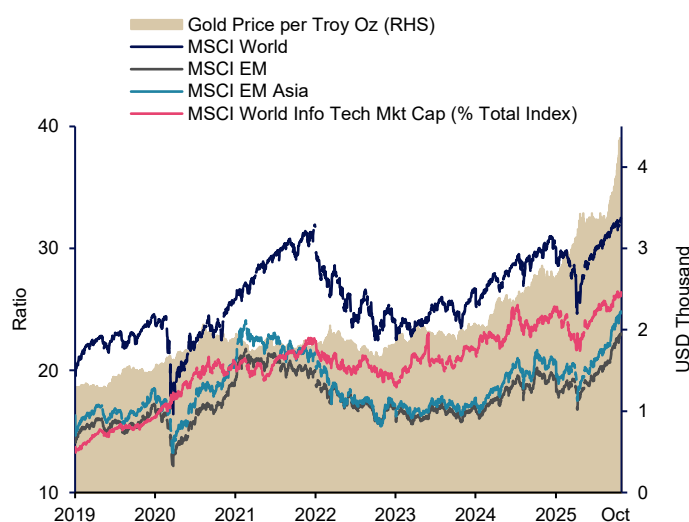
Public equity markets have rebounded to record highs, although the concurrent sharp rise in gold prices and unconventional movements in the US dollar suggest some unease about the trajectory of the markets (**Chart 1.3**). The continued divergence between equity market valuations and rising downside risks to growth raises the prospects of disorderly corrections in the event of shocks.

Much of the rise in equity markets has been driven by a surge in investments related to AI, potentially leaving many investors significantly exposed to the information technology sector. Some Big Tech firms (primarily hyperscalers) have also turned to the use of novel and potentially circular private financing arrangements to fund their expansions. These include the use of special purpose vehicles, private credit structures and novel accounting treatment that could mask leverage and increase funding dependencies. As a result, some AI firms face intensifying pressure to generate sufficient revenue to repay a growing base of public investors and private creditors.

A retrenchment of optimism in AI's ability to generate sufficient future returns may lead to sharp corrections in the broader equity market and further defaults in the private credit market. The materialisation of this risk could be exacerbated by the build-up of leverage, with spillovers to economic growth.

Chart 1.3 Equity valuations have risen in tandem with gold

MSCI equity indices (CAPE ratio) and Gold price



Source: Bloomberg, MAS estimates

Sovereign debt sustainability concerns have grown with widening fiscal deficits and rising term premia

Sovereign debt vulnerabilities have risen with widening fiscal deficits and already high public debt levels (**Chart 1.4**). Debt-servicing costs continue to increase alongside rising debt, with interest payments growing as a proportion of government revenues (**Chart 1.5**). As gaps between growth rates and interest rates have narrowed over the past two years (**Chart 1.6**),

public debt sustainability concerns could intensify should growth fail to keep pace with higher financing costs. Such concerns also constrain the use of fiscal policy to respond to any shocks.

Public debt sustainability concerns have resulted in a synchronous rise in bond term premia across AEs, leaving long-term yields relatively elevated despite declining policy rates. This will keep long-term borrowing costs high for governments. Shrinking demand for long-term bonds from institutional investors due to pension and regulatory reforms in some AEs, and from central banks unwinding their quantitative easing programmes, may exacerbate fiscal constraints.

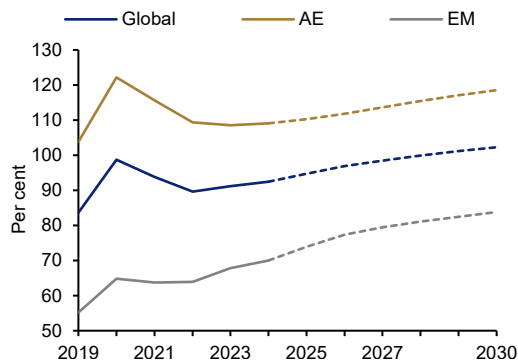
To manage borrowing costs, some governments have shifted bond issuance towards shorter maturities (**Chart 1.7**), reducing financing costs over the short term but then incurring more frequent refinancing cycles and greater rollover risks. These dynamics also raise concerns of fiscal dominance, as there could be greater pressure from governments for monetary easing, tolerance of stealth inflation or currency depreciation. Increased interdependency between monetary and fiscal policies could undermine central bank credibility and de-anchor inflation expectations. This could set off a destabilising spiral of higher inflation expectations and rising long-term yields, offsetting any relief from lower short-term borrowing costs.

Taken together, these risks raise the prospect of a disorderly adjustment in sovereign bond markets. Potential triggers could include episodes of political uncertainty, adverse growth or inflation surprises, or escalating geopolitical and trade tensions that drive reassessments of sovereign creditworthiness and risk premia. Market strains may be amplified by liquidity and leverage risks accumulating in hedge funds (**Chart 1.8**) and other NBFIs that have a growing footprint in financial market intermediation.

Beyond market disruptions, sudden increases in government bond yields can also result in significant mark-to-market losses in bank and NBFI balance sheets. A major repricing of sovereign credit, which typically serves as a benchmark for key lending rates, could spill over to the private sector, creating a feedback loop of higher borrowing costs and debt-servicing pressures that could weigh on growth.

Chart 1.4 Public debt is elevated and projected to rise further

Gross government debt as a share of nominal GDP

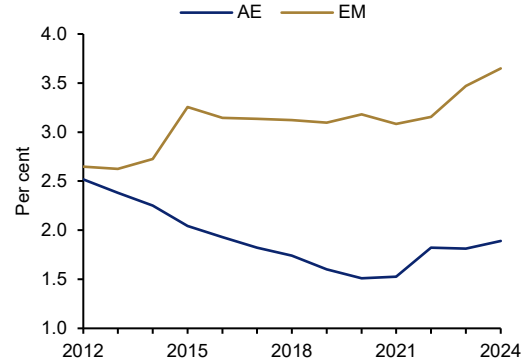


Source: IMF

Note: Dotted lines indicate projections by the IMF.

Chart 1.5 Sovereign interest payment burdens are rising

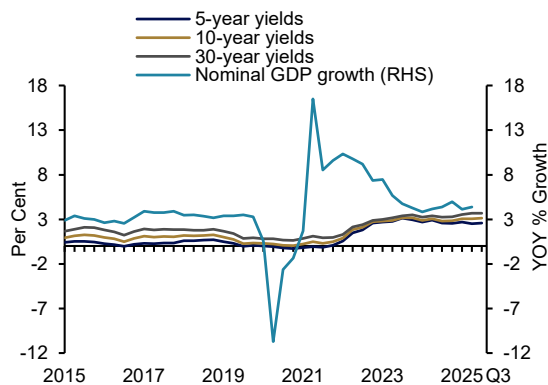
General government interest expense as a share of GDP



Source: IMF

Chart 1.6 Yields in AEs are rising and approaching nominal growth rates

Average sovereign bond yields and nominal GDP growth in AEs

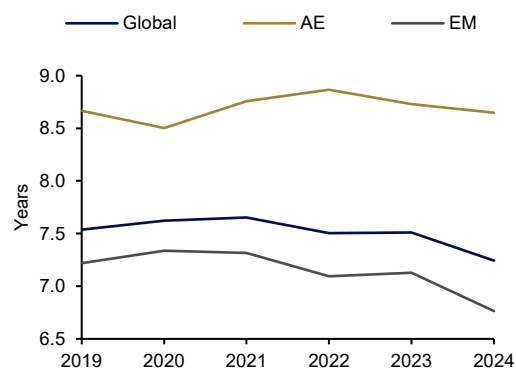


Source: Macrobond, Haver Analytics, MAS estimates

Note: Countries include the US, UK, Japan, Germany, Italy, France, Spain, Canada, Sweden, Netherlands, Denmark and Belgium.

Chart 1.7 Sovereign bond maturities are shortening

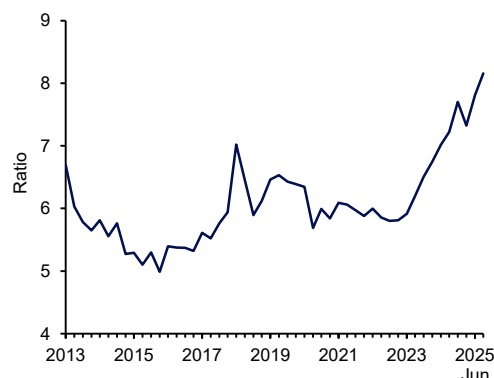
Average residual maturity on government bonds



Source: BIS, MAS estimates

Chart 1.8 Hedge funds' use of leverage continues to increase

Ratio of gross notional exposure to net asset value (NAV)



Source: US Securities and Exchange Commission, MAS estimates

Signs of credit stress are emerging in certain sectors

Banks globally have demonstrated resilience thus far with aggregate non-performing loan (NPL) ratios remaining stable (**Chart 1.9**), although loan-loss provisions have increased in some countries on expectations of higher credit losses. (**Chart 1.10**).

Banks in some AEs continue to report higher delinquency rates for their auto and credit card portfolios,² with some resurgence in student loan delinquencies. Although household finances remain healthy in aggregate (**Chart 1.11**) on the back of income growth, softer economic conditions and weaker labour markets could affect the financial resilience of the more vulnerable households.

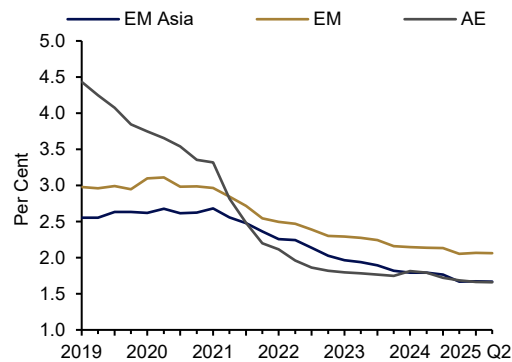
Credit to non-financial corporates (NFCs) has been stable, although corporate debt in EM Asia remains elevated relative to other regions (**Chart 1.12**). Firms have benefitted from favourable financing conditions, with corporate bond spreads at historically low levels. However, the current benign pricing of corporate credit risks could reverse sharply should trade-oriented corporates report lower profits or face cashflow strains.

At the same time, banks' growing links to NBFIs raise contagion, liquidity and credit risks. Some global banks have reported high-profile credit losses from distressed private credit-backed borrowers, provoking concerns about weaker credit underwriting and loan monitoring processes. These risks are amplified by private credit firms' opacity and use of complex financing structures involving leverage.

² See Board of Governors of the Federal Reserve System. 2025. "Financial Stability Report—April 2025."; Bank of Canada. 2025. "Financial Stability Report—2025."

Chart 1.9 NPL ratios remain subdued

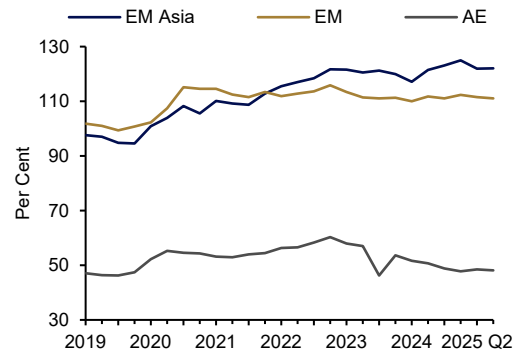
NPL ratio



Source: Haver Analytics, MAS estimates

Chart 1.10 Loan-loss provisions are rising in some EMs

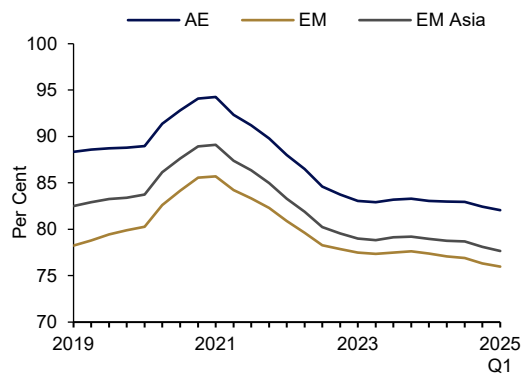
Provision coverage ratio



Source: Haver Analytics, MAS estimates

Chart 1.11 Household debt continues to decline

Household and Non-Profit Institutions Serving Households (NPISH) debt to GDP

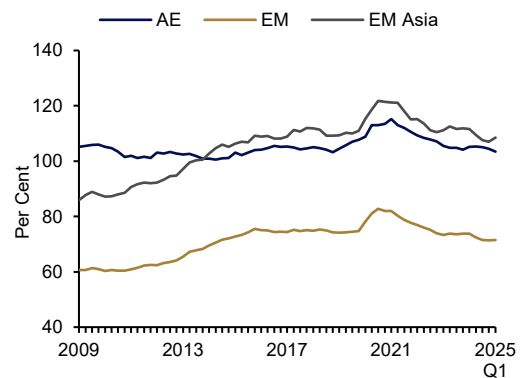


Source: Haver Analytics, MAS estimates

Note: Aggregates calculated using a simple average.

Chart 1.12 Corporate debt in EM Asia remains relatively elevated

Credit to NFCs as a percentage of GDP



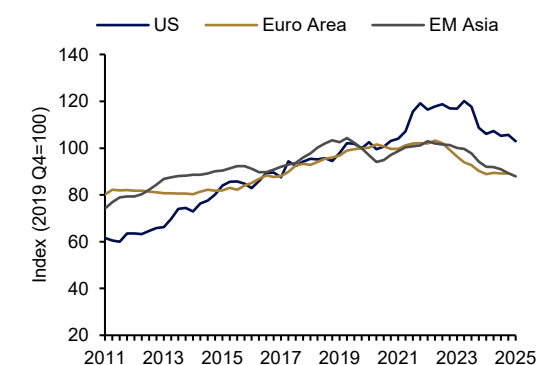
Source: Haver Analytics, MAS estimates

Note: Aggregates calculated using a simple average.

The commercial real estate (CRE) sector also remains a key source of credit risk amid an uneven global downturn. High vacancy rates continued to weigh on office valuations given persistent hybrid work trends (**Chart 1.13**). Considering banks' substantial exposure to the CRE sector, a sharp deterioration in loan quality would weaken bank balance sheets, with potential knock-on effects on credit availability to property and related sectors (**Chart 1.14**).

Chart 1.13 Prolonged downturn in global CRE sectors

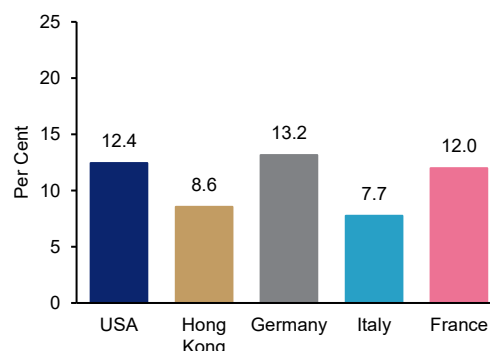
CRE price indices



Source: Haver Analytics

Chart 1.14 Banks in major jurisdictions have sizeable exposures to the CRE sector

Credit to the CRE sector relative to total bank credit



Source: Haver Analytics, CEIC, MAS estimates

Note: Figures for Germany, Italy and France are computed from real estate loans issued to NFCs.

Regional exporters face pressures from trade tariffs and dollar weakness

The USD has depreciated by 7% on a trade-weighted basis since the beginning of the year, although it has stabilised since July. For regional economies, USD weakness has limited the effectiveness of local exchange rates as a shock absorber in the event of a global growth slowdown. As regional trade is mostly invoiced in USD, exporters also face lower revenues in local currency terms. If expectations of USD weakness persist, exporters may be incentivised to increase FX hedging of their expected export revenue, adding to their costs.

The impact of appreciating regional currencies (**Chart 1.15**) is compounded by tariffs levied by the US. While real production and export volumes have been supported by the front-loading of demand in advance of tariff implementation, there is evidence of broad-based price weakness in the region as producers and exporters have cut prices (**Chart 1.16**). The resulting squeeze on profit margins could strain exporters' cashflows and constrain their ability to restructure supply chains, with ASEAN exporters in the Consumer Discretionary and Materials sectors particularly affected.³

Regional equity markets may also face a correction as weaker corporate earnings are priced in. Potential market volatility and a disorderly pullback in financing could amplify financial stress (see Box A "ASEAN Financial Stress Index" for more details on the construction of an index to measure financial stress and contagion in the region).

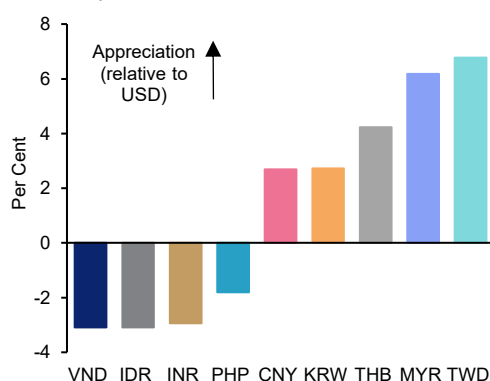
Regional economies remain susceptible to currency and capital flow volatility. There could be a reversal of USD depreciation and local currency appreciation trends once the effects of higher tariffs have passed through to trade and economic growth in the region. Against slower growth and a possible resurgence in local currency weakness and capital outflows, regional central banks will face complex trade-offs between price stability and external stability.

³ Findings are based on a stress test conducted on ASEAN listed corporates.

Regional central banks have acted pre-emptively to mitigate risks, building up FX reserves and easing rates in the current period of USD weakness without compromising external stability. Most have also deployed expanded policy toolkits, including macroprudential policy and capital flow measures to temper exchange rate and capital flow volatility. EMs' relative resilience through recent shifts in the global financial cycle underscores the importance of credible macroeconomic policy responses and adequate FX reserves in safeguarding financial stability (see Special Feature 1 "Exchange Rate Flexibility in Asia and Latin America" for an analysis of different policy interventions by EM central banks).

Chart 1.15 Many Asian currencies have appreciated amid broad-based dollar weakness

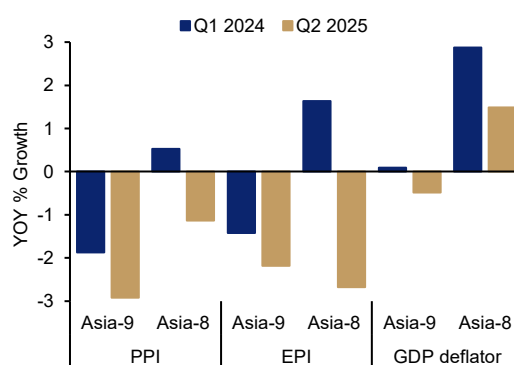
YTD currency movements



Source: Bloomberg, MAS calculations

Chart 1.16 Multiple indices point to broad-based price weaknesses in the region

Regional price indices, weighted by real GDP



Source: Haver Analytics, CEIC, MAS calculations

Note: Asia-9 refers to China, India, Indonesia, Malaysia, Philippines, South Korea, Taiwan, Thailand and Vietnam. Asia-8 includes the same countries, excluding China. EPI = Export price index; PPI = Producer price index

Emerging global vulnerabilities have increased and warrant close monitoring

Rising protectionism and trade tensions, along with continuing geopolitical conflicts, have intensified concerns about global economic and financial fragmentation. A disorderly shift towards a multi-polar world could unravel cross-border trade and investment flows, potentially reducing liquidity and investment diversification in the global financial system, and impeding global economic activity. The costs are unlikely to be evenly distributed, with economies that lack deep or integrated capital markets most affected, reinforcing structural imbalances from asymmetric access to international financial markets, higher borrowing costs, and greater vulnerability to capital outflows.

Meanwhile, the global payments landscape is being reshaped by the growing adoption of stablecoins. While stablecoins may offer innovation and efficiency gains, they also have implications for monetary and financial stability. Stablecoins have the potential to impact the cost and composition of bank funding, while issuers' reserve holdings of traditional financial assets may heighten contagion risks in the broader financial system, especially for banks that hold significant stablecoin reserves as deposits. Notwithstanding the benefits that technology might provide, stablecoins may also be a channel for the transmission of stresses from crypto-

asset trading and decentralised finance activities to the rest of the financial system, more so if such activities are intermediated by lightly regulated or unregulated entities. Accordingly, several jurisdictions are examining the monetary and financial stability implications of stablecoins and putting in place appropriate regulatory frameworks to address potential risks.

Box A

ASEAN Financial Stress Index

A high-frequency composite Financial Stress Index (FSI) for ASEAN measures regional financial stress

The ASEAN FSI is a composite measure of regional financial stress. It aggregates the movements of key ASEAN financial markets (e.g. equity, FX, bond, short-term funding markets) and captures correlations to detect potential financial contagion across asset markets and jurisdictions.

By capturing systemic risk dynamics in a consistent manner, it supplements surveillance of individual asset and funding markets. Idiosyncratic stresses in specific financial markets will result in a spike in the ASEAN FSI only if they are found to correlate with other market movements. An overall increase in market co-movements indicates a higher risk of financial contagion.

Decomposing the movements of the ASEAN FSI into the relative contributions from different markets can help to identify the sources of financial stress. The ASEAN FSI also facilitates assessment of the severity of current financial market stress by comparing against its historical levels.

Overall, the FSI helps to augment slower-moving indicators such as changes in credit conditions and economic activity levels, particularly when estimating the impact of disruptions in the financial cycle.

Methodology

The ASEAN FSI is constructed from 15 financial market variables across four key markets—the equity, FX, bond and short-term funding markets of each of the ASEAN-5 economies⁴—as well as volatility indices from global financial markets. These selected variables capture some of the features of financial stress—including increased uncertainty, greater information asymmetry and reduced willingness to hold riskier and less liquid assets.⁵

The FSI's indicators include variables expressed as first order moments (e.g. change in prices relative to a year ago) and second order moments (e.g. volatility of returns). The first order moments reflect protracted deviations from historical prices and valuations, while the second order moments capture volatility spikes which quickly appear during episodes of market duress, and are thus useful as high frequency signals.

For each of the five markets, the data is standardised by ranking each indicator against historical trends on a rolling basis and assigning a score between zero and one based on these

⁴ This includes Indonesia, Malaysia, Philippines, Thailand and Vietnam.

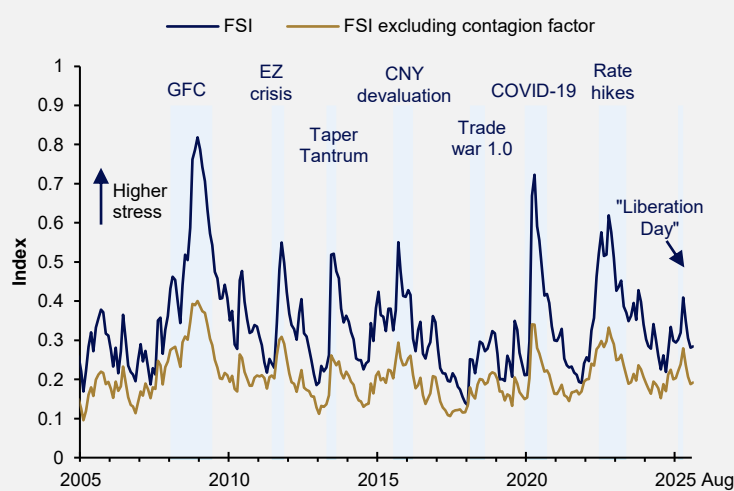
⁵ For more details, see Hakkio, Craig S., and William R. Keeton. 2009. "Financial Stress: What Is It, How Can It Be Measured, and Why Does It Matter?" *Economic Review* 94 (2): 5–50.

ranks. Subsequently, principal component analysis is used to extract the most important factor for each market segment.

Initially, equal weights are assigned to each of the asset or funding markets to compute a hypothetical FSI without contagion. Thereafter, by incorporating time-varying cross-market correlations in the computation, the final FSI captures the impact of cross-market contagion to the FSI during periods of stress (**Chart A1**).

Chart A1 The ASEAN-FSI successfully detected historical episodes of high financial stress

ASEAN FSI



Source: Bloomberg, MAS estimates

Note: Chart A1 shows both the FSI and FSI (no contagion), with the difference between the two lines being the contagion or correlation factor. The ASEAN FSI readings range from 0 to 1 (with 1 indicating the highest stress level).

The ASEAN FSI identifies previous financial stress episodes

The ASEAN FSI was back-tested against two stress episodes—the Global Financial Crisis (GFC, 2008 to 2009) and the COVID-19 pandemic (2020)—which are yardsticks for extreme stresses (**Chart A1**). The GFC and COVID-19 shocks registered peak FSI readings of 0.85 and 0.75 respectively, and these can serve as indicative thresholds for future financial stresses.

In addition, MAS also separately conducted empirical regressions to derive indicative crisis thresholds. These logistic regressions show that each standard deviation increase (i.e. one, two and three standard deviation) in the FSI from its historical mean is associated with a 12–21 percentage point increase in probability of being in a historical financial stress episode (**Table A1**).

- Financial stress is defined as periods where policy interventions are necessary to address systemic concerns arising from troubled financial institutions (FIs) or financial markets.
- In line with this definition, MAS deployed large language models (LLMs) to compile a series of policy interventions by global and regional fiscal and monetary authorities

for the period from 2005 to 2025. Policy interventions are used to identify financial stress episodes.

- By regressing the daily lagged FSI against a daily binary indicator of financial stress episodes (with an average 20-business day window), MAS derived statistically significant estimates—an increase in FSI indicates a higher probability of financial stress events as defined by authorities' policy interventions.

Table A1 Each standard deviation increase in the ASEAN-FSI from its historical mean indicates a higher probability of a financial stress episode

Logistic regression results

FSI levels	Standard deviation from mean	Percentile	Probability of financial stress	Increase in probability
0.34 [^]	0	59.8	11%	-
0.48	1	86.4	23%	12%
0.61	2	95.3	41%	18%
0.74	3	97.7	62%	21%

Source: Bloomberg, Information on historical financial stress episodes compiled using LLMs, MAS estimates

[^] Historical mean of FSI

Spikes in the ASEAN FSI are historically associated with lower domestic economic activity

MAS' analysis also found that higher FSI levels are associated with a decline in overall economic activity one month later in the historical sample period. In particular, the increase in the country-specific FSI above critical thresholds ranging from 0.41 to 0.52 is associated with a drag on overall economic activity. This suggests that the FSI could serve as an early-warning signal of a potential decline in real economic activity when financial stress thresholds are breached for a prolonged period.

- By regressing lagged values of the country-specific FSI⁶ against their respective Composite Economic Indices (CEI),⁷ the results show that a higher FSI reading one month prior is associated with a statistically significant decline in the current month's CEI.

⁶ In addition to an overall ASEAN FSI, country-specific FSI were estimated using the same set of 15 financial market variables for the respective countries.

⁷ This regression was performed for three ASEAN countries, namely Indonesia, Malaysia and Thailand. The CEI spans a wide range of economic activities (e.g. industrial output, consumption, trade, employment), thereby allowing for a broad coverage of the economy while reducing idiosyncratic noise of a single economic series. The regression specification also includes lagged values of the CEI as control variables.

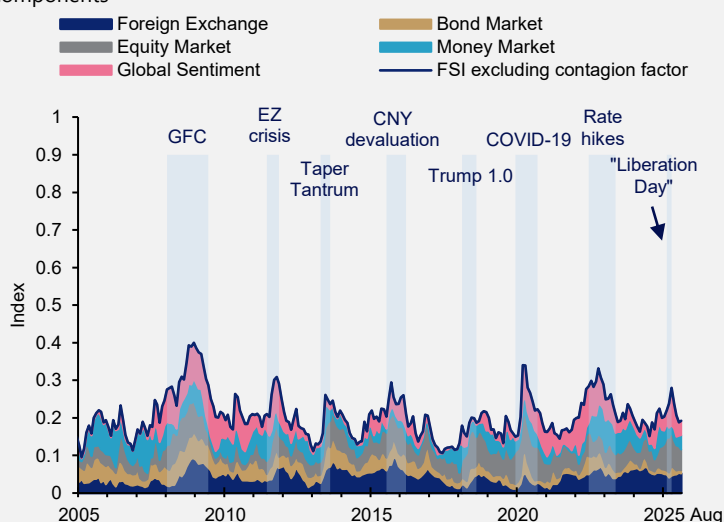
- To derive this critical threshold, MAS used threshold regression techniques⁸ to differentiate between a low and high financial stress regime. Under the low stress regime, lagged FSI values did not precede negative economic impacts. Conversely, under a high stress regime, the regressions identified a statistically significant negative impulse to the economy from high financial stress levels.
- The critical threshold at which economic effects become significant ranges from 0.41 to 0.52. For reference, the median FSI levels for these countries range from 0.28 to 0.31 during the historical sample period.

The April 2025 “Liberation Day” episode triggered a smaller uptick in the ASEAN FSI relative to the GFC and COVID-19 crises

Relative to the sharp spike in the ASEAN-FSI observed during the GFC and COVID-19 crises, the “Liberation Day” episode saw a smaller rise, with an average FSI reading of 0.41 over the month of April, which falls in the 81st percentile of historical FSI readings. The probability of a financial stress event⁹ rose to more than 20%. However, the surge in the FSI saw a relatively swift recovery to more stable levels due to a quick resolution in global financial market volatility, relatively stable domestic bond markets and a recovery in regional equity prices (**Chart A2**).

Chart A2 Decomposing the FSI into its subcomponent markets allows for the identification of the sources of financial stress

ASEAN FSI sub-components



Source: Bloomberg, MAS estimates

⁸ Specifically, two approaches were employed to estimate the critical threshold: (i) a threshold regression approach, where two separate sets of coefficients are estimated for the high and low stress regimes. These regimes are differentiated by the FSI at the critical threshold; (ii) a linear spline regression approach, where one set of regression equations are estimated, but with a kink at the critical threshold.

⁹ As before, a financial stress event is defined as a period with policy interventions from authorities to address systemic concerns arising from troubles in FIs or financial markets.

2 Singapore Financial Conditions

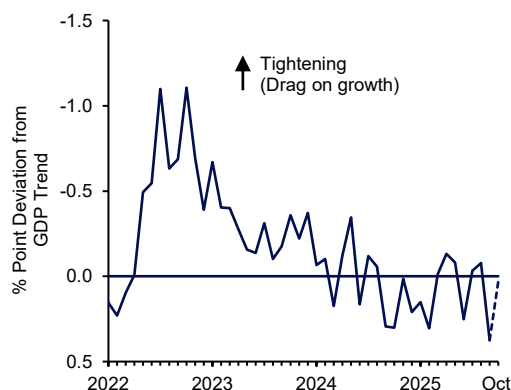
Domestic financial conditions have been supportive of growth

Financial conditions in Singapore¹⁰ have been mildly accommodative over the past year (**Chart 2.1**), supported by strong tailwinds from buoyant global financial markets and more accommodative monetary policy by major central banks. Domestic bond and equity markets have both registered substantial gains, with long-term yields falling 102 basis points (bps) and the Straits Times Index (STI) rising 17% since the beginning of the year to reach an all-time high.

Interest rates in Singapore declined significantly, with the 3-month SORA falling from 3.59% in Q3 2024 to 1.72% in Q3 2025, while investment grade credit spreads continued to compress (**Table 2.1**). Bank lending to consumers and businesses expanded for the second straight year, growing 6.2%, and supporting an 9.5% increase in the money supply, as measured by M2. The growth in money supply was further supported by strong growth in SGD deposits by resident individuals and non-financial corporates.

Chart 2.1 Domestic financial conditions have been mildly supportive over the past year

Domestic Financial Conditions Index (FCI)



Source: Bloomberg, Department of Statistics (DOS), Urban Redevelopment Authority (URA), MAS estimates

Note: The Oct 2025 FCI value is an interim figure based on latest available component data. The FCI methodology has been enhanced since the 2024 FSR.

Table 2.1 Key indicators of domestic financial conditions

	Q3 2023	Q3 2024	Q3 2025
Domestic FCI (% point deviation from GDP trend)	-0.20	0.04	0.09
3-month compounded SORA (%)	3.69	3.59	1.72
10-year SGS yields (%)	3.15	2.82	1.95
STI (points)	3,239.2	3,443.8	4,222.8
CEMBI Investment Grade+ SG Spread*	133.8	78.8	61.3

Source: Bloomberg, J.P. Morgan Markets, MAS

Note: Figures are quarterly averages.

* This measure is computed by J.P. Morgan Markets as the yield spread of USD corporate bonds over US Treasury bonds.

Financial markets have been resilient

Domestic markets continued to function normally despite increased uncertainty in the global environment. Spreads remained tight. Singapore's FSI, an indicator of stress and contagion, rose sharply in April 2025 after the announcement of US reciprocal tariffs (**Chart**

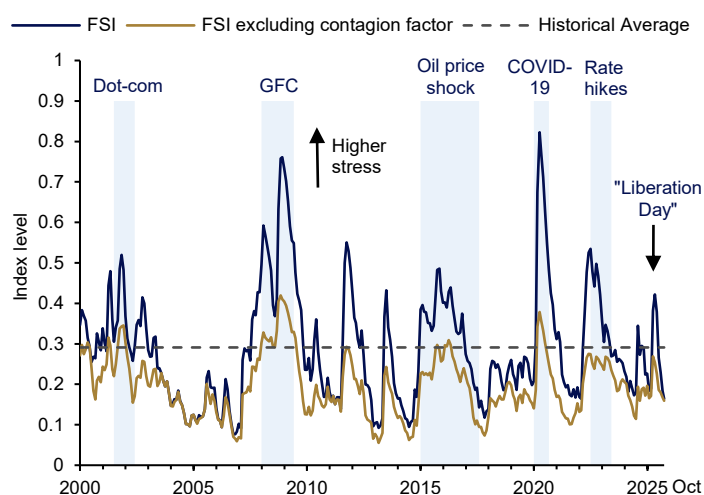
¹⁰ For more details on the FCI, please see "Special Feature 2: Expanding the Toolkit for Macroprudential Surveillance" in the 2024 FSR.

2.2), reflecting heightened levels of volatility in global financial markets. However, the increase in risk aversion was short-lived and the level of stress has since eased considerably.

While Singapore's FSI has moderated from its peak in April to fall below its long-term average, renewed trade conflict or an escalation in global fiscal sustainability concerns could trigger revisions in investor risk sentiment. This could lead to potentially sharp corrections in international financial markets, especially amid relatively rich valuations and compressed risk premia.

Chart 2.2 Market stress induced by "Liberation Day" tariff announcements subsided quickly

Domestic FSI



Source: Bloomberg, MAS estimates

Singapore's FIs flagged macroeconomic uncertainty and geopolitical risk as key risks

Singapore's FIs cited macroeconomic uncertainty as the most impactful risk to the domestic financial system in the most recent MAS Systemic Risk Survey. Concerns over geopolitical risk were also widely expressed, with respondents identifying geopolitical conflicts and trade tensions as key themes. The findings are discussed in more detail in Box B "Systemic Risk Survey of Singapore's Financial Sector".

Box B

Systemic Risk Survey of Singapore's Financial Sector

Introduction

The chief risk officers of key FIs in Singapore highlighted geopolitical risk and macroeconomic uncertainty as the two most significant risk factors for Singapore's financial system, according to the 2025 Systemic Risk Survey (**Table B1**).

Table B1 FIs' perceptions of risks to Singapore's financial system

Top 5 risk categories cited by respondents	% of respondents		Impact ranking	
	March 2025	September 2025 (Δpp)	March 2025	September 2025 (Δ)
Macroeconomic uncertainty (e.g. growth slowdown and inflation resurgence)	43	59 (+16)	3	1 (12)
Geopolitical risk induced by escalation of ongoing conflicts, as well as trade tensions and uncertain policies of newly elected governments	80	77 (-3)	1	2 (11)
Concentration risks relating to reliance on key third-party vendors	21	32 (+11)	4	3 (11)
Technology and cyber risk	68	75 (+7)	2	4 (12)
Emerging technologies (e.g. AI, ML, crypto/stablecoins, quantum computing)	29	34 (+5)	5	5

Source: MAS estimates

Note: (i) Impact ranking is determined by the average of respondents' ranking of each risk by potential impact; (ii) both runs of the survey had response rates of 98% from 57 FIs.

Uncertainty over the economy increased as trade tensions escalated

Citations of macroeconomic uncertainty as the most impactful risk increased markedly between the H1 and H2 surveys, as the boost from front-loading activities faded and higher tariffs took effect in the region, reducing growth prospects. Respondents also expressed concerns over fiscal sustainability in AEs, noting that upward pressure on long-term bond yields could weaken the transmission of monetary easing and result in higher borrowing costs. Respondents also noted that USD weakness could lead to losses on unhedged holdings of US assets and increased hedging activity, leading to higher costs for exporters and investors.

Concerns over geopolitical risk were widely shared

Geopolitical issues were the most-cited risk this year, with respondents identifying two prominent themes—conflicts and trade tensions.

FIs cited the protracted conflicts in Eastern Europe and the Middle East, along with potential US-China frictions, as areas of concern. Many noted how these conflicts might trigger energy and commodities supply shocks. Nearly a third of FIs citing geopolitics mentioned sanctions as a key risk of conflict escalation. Sanctions could affect banks' and fund managers' operations by making compliance more costly, compelling the abrupt sale of assets that could lead to financial market volatility.

Beyond near-term credit and market risks, FIs were also wary of the persistent effects of trade tensions. They noted that geopolitical factors increasingly shape supply chain configurations, particularly in strategic sectors like technology and critical raw materials. Against these shifts, FIs will need to reassess risks around supply chain financing, as well as corporate and sectoral exposures.

Other notable risks centred around concentration in emerging technologies

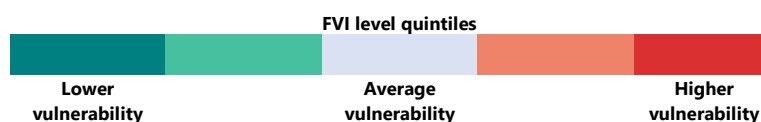
Dependence on a limited number of third-party IT vendors (particularly for cloud computing and IT security services) continues to be a concentration risk worry for FIs, which acknowledge that an interruption of services provided could have a severe impact on their operations. Concerns about the advent of quantum computing and the ability of the financial sector to adapt adequately to accompanying challenges were also noted.

FIs recognised potential risks related to AI, including heightened risks of cyber-attacks and scams that leverage the use of AI. FIs also voiced unease over greater adoption of stablecoins and their integration with the broader financial system. Mainstream adoption of stablecoins could affect banks' funding stability and raise borrowing costs. Stablecoin issuers' holdings of short-term sovereign bonds as reserves also establishes a direct link from stablecoins to traditional financial markets, where large and abrupt redemption shocks may involve fire sales of reserve assets, disrupting core funding markets and amplifying market volatility.

3 Singapore Corporate Sector

3.1 Assessment of vulnerabilities

Corporate sector Financial Vulnerabilities Index (FVI) ¹¹ level	Q2 2024	Q2 2025
Overall corporate FVI		
Leverage risk		
Liquidity risk		
Maturity risk		
Foreign currency risk		

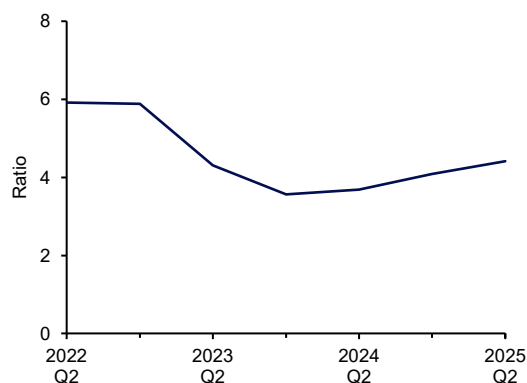


Singapore corporates have benefitted from the resilient domestic growth momentum and the recent easing in financing conditions, which has enabled them to strengthen their financial position over the past year. Leverage risk has remained contained, with improvements in debt-servicing capacity (**Chart 3.1**) underpinned by resilient corporate earnings and a more accommodative financing environment (see Chapter 2 “Singapore Financial Conditions”). The earlier strains from higher borrowing costs have eased (**Chart 3.2**), alongside the decline in the corporate debt-to-GDP ratio from 122% in Q2 2024 to 116% in Q2 2025.

¹¹ The corporate sector FVI consists of four risk indicators: (i) leverage risk measures firms’ debt levels and debt repayment capabilities, (ii) liquidity risk assesses the adequacy of firms’ liquid assets to cover short-term liabilities, (iii) maturity risk gauges firms’ ability to rollover short-term debt, and (iv) foreign currency risk is proxied by the share of foreign currency debt to total debt. The corporate FVI has been enhanced since the FSR 2024 to improve the scope of firms included in the computations.

Chart 3.1 Firms' debt-servicing capacity has improved since last year

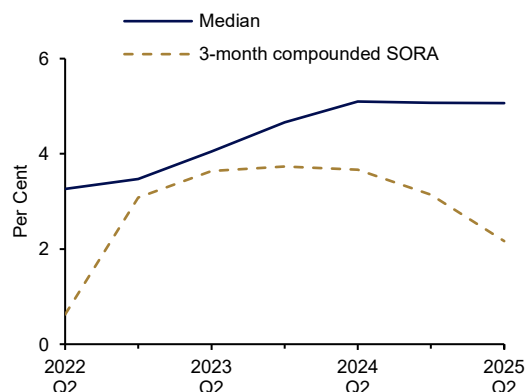
Median interest coverage ratio (ICR) of SGX-listed firms



Source: LSEG Workspace, MAS estimates

Chart 3.2 Borrowing costs have stabilised

Interest expense as a share of total debt of SGX-listed firms



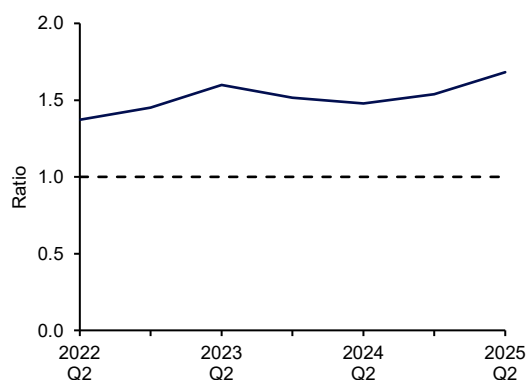
Source: LSEG Workspace, MAS estimates

Firms have maintained healthy liquidity buffers and debt maturity profiles. Strong earnings have enabled firms to build up ample cash buffers, with cash cover ratios improving in recent quarters (**Chart 3.3**). Debt maturity profiles have stayed stable with the ratio of short-term debt to total debt holding steady (**Chart 3.4**). Bonds maturing by the end of 2026 comprise a manageable 16% of total outstanding corporate bonds as of Q3 2025.

Foreign currency risk has also not risen over the past year. Since 2018, foreign-currency-denominated bonds have accounted for a stable 60% of total outstanding corporate bonds in Singapore. Financial disclosures lodged with Singapore Exchange (SGX) show that firms that borrow in foreign currencies tend to either be naturally hedged, with revenues in currencies matching their liabilities, or actively hedged through derivatives.¹²

Chart 3.3 Firms continued to hold sufficient cash to repay short-term debt at short notice

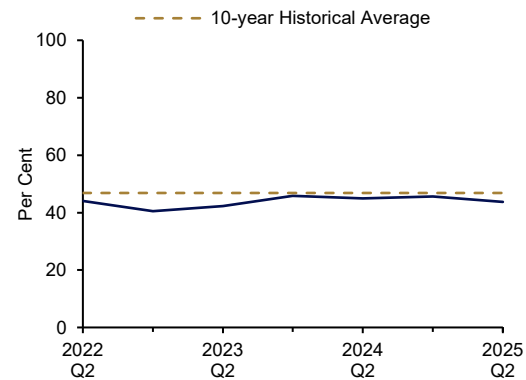
Median cash to short-term debt of SGX-listed firms



Source: LSEG Workspace, MAS estimates

Chart 3.4 Short-term debt to total debt ratios have remained stable

Median short-term debt to total debt ratio of SGX-listed firms

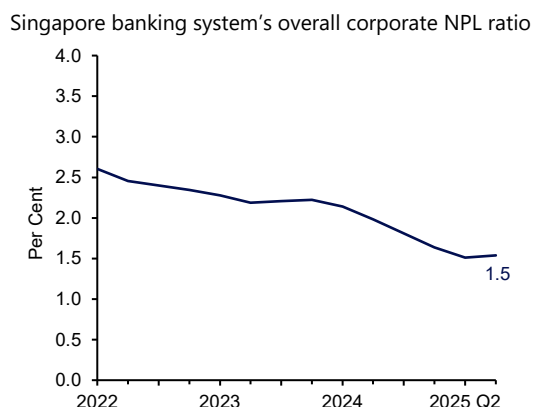


Source: LSEG Workspace, MAS estimates

¹² Based on an analysis of financial disclosures of more highly leveraged Singapore-listed firms.

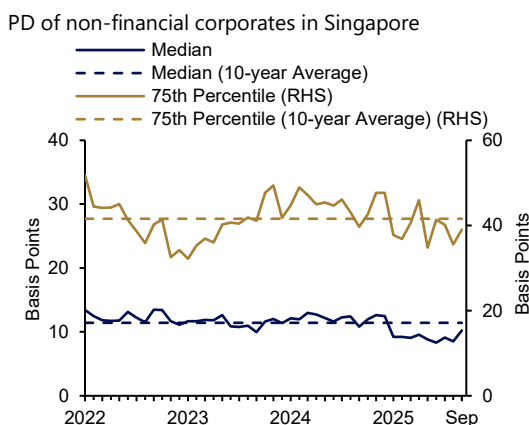
Total defaults, including those of small and medium-sized enterprises (SMEs), have remained low (**Chart 3.5**). The SME NPL ratio has been declining since 2024 (see Chart Panel 3A “Small and Medium-sized Enterprise Financing Conditions”, Chart 3A5), indicating that SME credit quality has been relatively stable amid the ongoing runoff in SME loans under government schemes (see Box C “SMEs’ Access to Financing”). The National University of Singapore Credit Research Initiative (NUS-CRI)’s Probability of Default (PD) indicator also suggests that near-term corporate credit risks remain contained, as PDs of listed firms—including those of more vulnerable firms—have trended lower in 2025 (**Chart 3.6**).

Chart 3.5 Singapore banking system’s corporate NPL ratio has stayed low in 2025



Source: MAS

Chart 3.6 Probability of default of Singapore firms has remained below historical averages



Source: NUS-CRI, MAS estimates

3.2 Outlook

Corporate debt-servicing capabilities should remain more than sufficient in the context of a more accommodative global interest rate environment and benign domestic financial conditions. That said, the business outlook is subject to downside risks amid macroeconomic uncertainties and trade tensions, reflected in the softening of business sentiment as revenue and profit expectations have weakened (**Chart 3.7**).

To evaluate the corporate sector’s resilience to tariff shocks and higher credit costs, MAS conducted a stress test on SGX-listed corporates by applying market-implied firm-level revenue shocks¹³ accompanied by a 100 bps rise in interest rates.¹⁴ The stress test was calibrated to be more severe than the FSR 2024 exercise, with more conservative firm-level differentiated assumptions around revenue outcomes,¹⁵ to assess both debt-servicing capacity as well as cashflow adequacy. The results suggest that most corporates are resilient

¹³ Market-implied firm-level revenue shocks were calibrated based on firm-level equity price movements and volatility from tariff related shocks. The revenue declines averaged 10% across firms and ranged up to 50%.

¹⁴ A simulation of a larger credit risk premium charged by banks despite the more accommodative interest rate environment.

¹⁵ The 2025 stress test assumed firm-specific revenue shocks while holding a proportion of firms’ costs fixed, thereby resulting in a greater impact on earnings compared to the FSR 2024 exercise, which assumed costs fell by the same proportion as revenues.

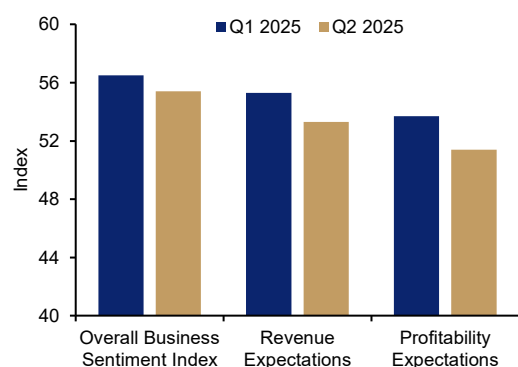
to such shocks, with cash reserves providing adequate buffers. Based on their Q2 2025 financial positions, the percentage of firms-at-risk¹⁶ stood at 25% after the heightened stress. Debt-at-risk, i.e. debt held by firms-at-risk, came in lower at 12% of overall corporate debt (**Chart 3.8**).

Corporates that are less resilient under stress with more debt-at-risk are predominantly externally-oriented firms that are more vulnerable to a resurgence in global trade tensions, as well as those with capital-intensive business models and higher operating leverage. Smaller firms also comprise a disproportionate share of firms-at-risk, due to their lower profit margins and thinner buffers.

Amid the macroeconomic uncertainties, more vulnerable corporates and those at risk should ensure that their buffers are adequate, including by managing their debt obligations prudently and planning for increases to liquidity buffers.

Chart 3.7 Business sentiment, revenue and profit expectations eased in Q2 2025

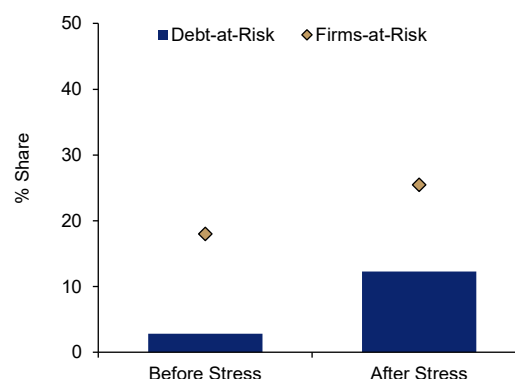
Singapore Business Federation's national business survey



Source: Singapore Business Federation

Chart 3.8 Most Singapore corporates would remain resilient under adverse shocks

Listed firms with either ICR<1 or negative cashflow with <6 months cash buffer



Source: LSEG Workspace, MAS estimates

¹⁶ Defined conservatively as either having ICR of less than one or negative cashflows with cash buffers that cover less than six months of the shortfall.

Box C

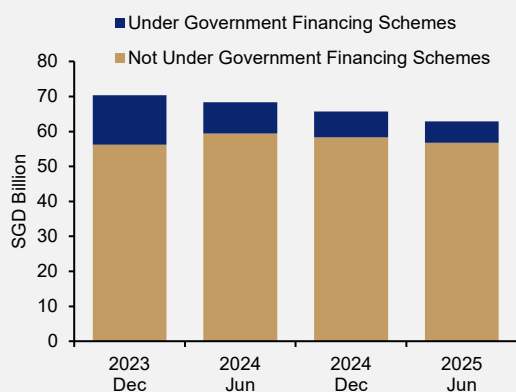
SMEs' Access to Financing

Over the past year, credit extended to SMEs has continued to ease at a moderate rate (see Chart Panel 3A, Chart 3A1). The decline mainly reflected the natural runoff of loans following the expiry of the enhanced Enterprise Singapore (ESG) loan schemes for SMEs introduced during the COVID-19 pandemic. By contrast, outstanding loans to SMEs not under ESG loan schemes remained steady, suggesting that underlying credit supply to SMEs has broadly held up (**Chart C1**).

In fact, the overall number of SME borrowers has increased (see Chart Panel 3A, Chart 3A2), while average interest rates on new SME loans have shifted downward alongside the fall in the SORA (**Chart C2**). Loan volumes for micro-enterprises, which are typically more vulnerable when financial conditions tighten, have also remained stable (**Chart C3**).

Chart C1 Outstanding loans not under government schemes has remained steady

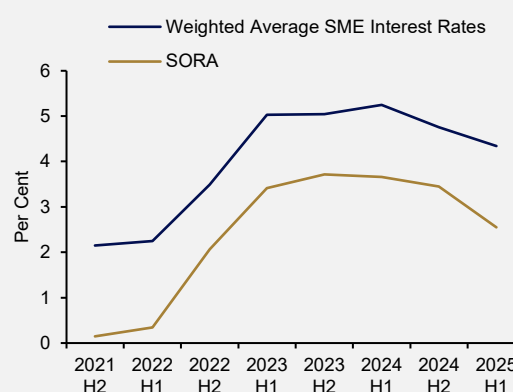
Outstanding loans to resident SMEs



Source: MAS survey

Chart C2 Average interest rates have declined alongside the fall in risk-free rates

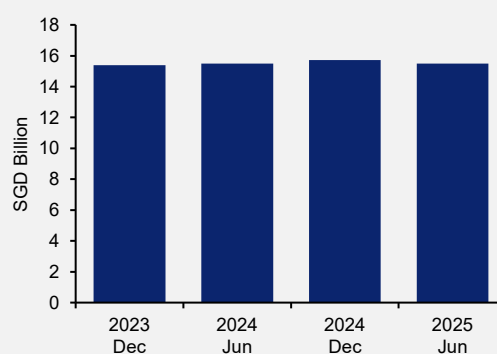
Overall SME interest rates charged on new loans



Source: MAS survey

Chart C3 Micro-enterprises have seen stable loan volumes

Outstanding loans to micro-enterprises

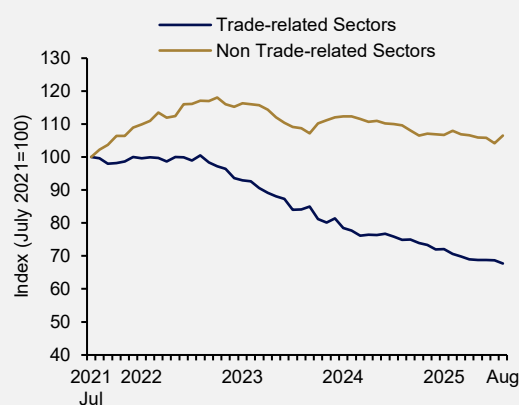


Source: MAS survey

Lending to a sub-set of SMEs in trade-related sectors has been declining (**Chart C4**). SMEs in the trade-related sectors have weaker profitability compared to their large corporate counterparts, thus constraining their ability to absorb revenue losses from shocks (**Chart C5**). In contrast, the decline in loans to non-trade-related SMEs over the past year was a continuation of the normalisation following an earlier surge in lending.

Chart C4 Loans to trade-related sectors have been declining

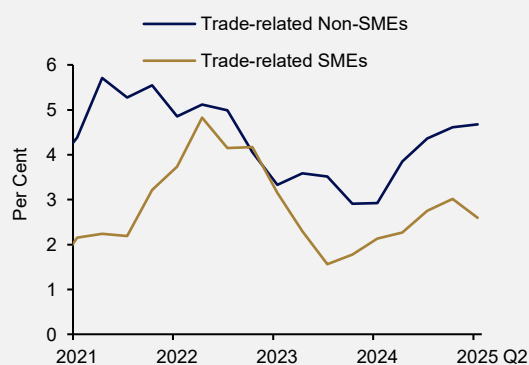
Outstanding SME loans



Source: MAS

Chart C5 Trade-related SMEs are generally less profitable than their large corporate counterparts

Median net profit margin of listed trade-related firms



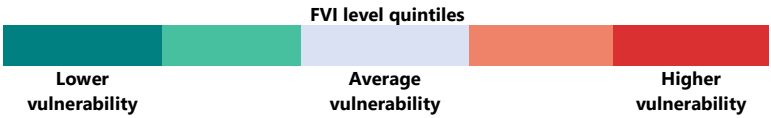
Source: LSEG Workspace, MAS estimates

The uncertain macroeconomic environment could weigh on prospects of SMEs. While the NPL ratio for the SME portfolio remains low (see Chart Panel 3A, Chart 3A5), banks are watchful of risks arising from shifts in global trade policies, and are prepared to engage and explore bilateral assistance measures with SME borrowers affected by tariff-related challenges. The Singapore Economic Resilience Taskforce (SERT) is also closely monitoring the impact on businesses and stands ready to provide support if necessary.

4 Singapore Household Sector

4.1 Assessment of vulnerabilities

Household sector FVI ¹⁷ level	Q2 2024	Q2 2025
Overall household FVI		
Leverage risk		
Liquidity risk		
Asset price volatility risk		



Overall household sector vulnerabilities remain low. Despite a modest rise due to an increase in outstanding mortgage loans, household leverage risk is assessed to be contained as household financial assets have grown faster than household debt. In Q2 2025, households’ financial assets were 8.5% higher than in the same period the year before, while overall household debt increased by only 6.1%. Income growth has also kept pace, keeping the household debt to personal disposable Income (PDI) ratio stable at around 1.1 (**Chart 4.1**). The increase in debt was led by mortgage loans (**Chart 4.2**), that are secured by property as collateral, and has been in line with a pick-up in transaction activity in the property market (see Box D “Update on the Private Residential Property Market”).

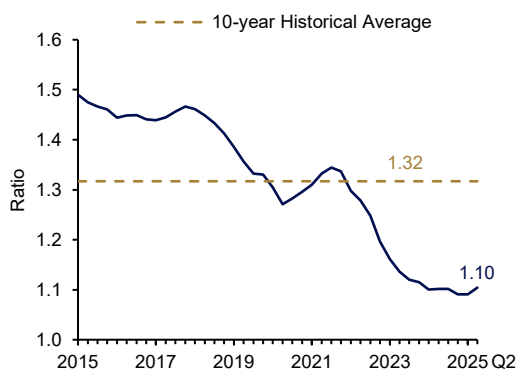
Debt-servicing capacity has remained healthy, underpinned by stable income growth¹⁸ and lower mortgage rates. Reflecting the decline in the SORA over the past year, borrowers across all income profiles who refinanced into lower rate packages have seen their mortgage servicing to income ratios fall—the median mortgage rate for newly originated loans has fallen by around 90 bps since Q2 2024. Accordingly, the credit quality of housing loans extended by FIs has also remained strong, with the NPL ratio at a low of 0.3% as of Q2 2025 (**Chart 4.3**).

¹⁷ The household sector FVI consists of three indicators: (i) leverage risk assesses household mortgage debt levels alongside debt repayment risk; (ii) liquidity risk assesses household vulnerability to short-term debt, such as credit card borrowings, compared to income and liquidity buffers; (iii) asset price volatility risk assesses households’ exposures to market risks in the private residential property market.

¹⁸ Moving four-quarter PDI recorded 5.9% y-o-y growth in Q2 2025, up slightly from 5.6% in Q2 2024.

Chart 4.1 Household debt-to-PDI has remained low in Q2 2025

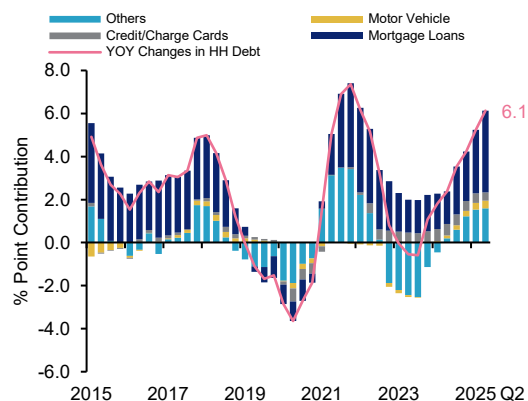
Household debt-to-PDI in Singapore



Source: DOS, MAS estimates

Chart 4.2 Increase in household debt mirrored the growth in mortgages

Contribution to the growth in household debt

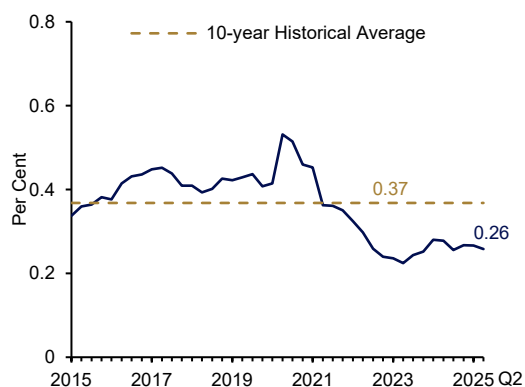


Source: DOS, MAS estimates

Liquidity risk for households has remained low as the growth in cash and deposits kept pace with that of shorter-term household liabilities. Further, the ratio of personal loans to PDI has been broadly stable at a low of around 0.3 over the past year (**Chart 4.4**). For credit card loans, rollover balances as a share of PDI edged up to 2.5% in Q2 2025, extending the trend increase since 2022 (**Chart 4.5**). Even so, a large majority of borrowers continues to be able to service their credit and charge card debts as delinquency rates have kept low at 0.9% (**Chart 4.6**).

Chart 4.3 NPL ratio on housing loans remained low in Q2 2025

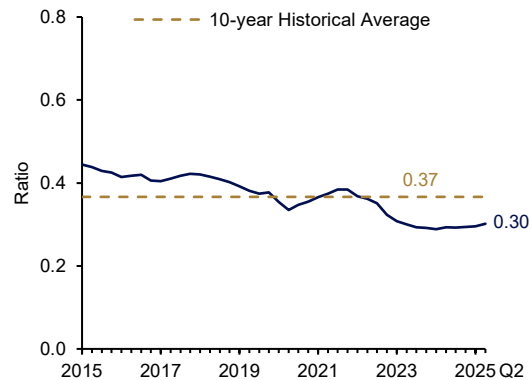
Housing NPL ratio



Source: MAS estimates

Chart 4.4 Personal loans to PDI ratio has been stable over the past year

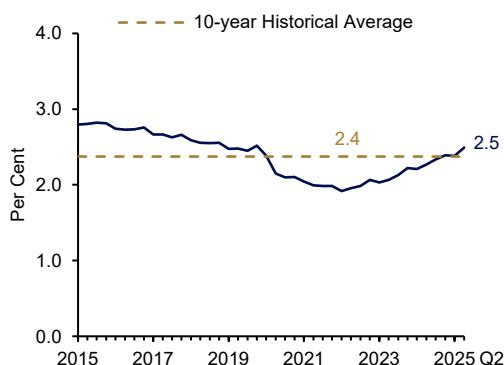
Personal loans to PDI



Source: DOS, MAS estimates

Chart 4.5 Rollover balances as a share of PDI has increased since 2022

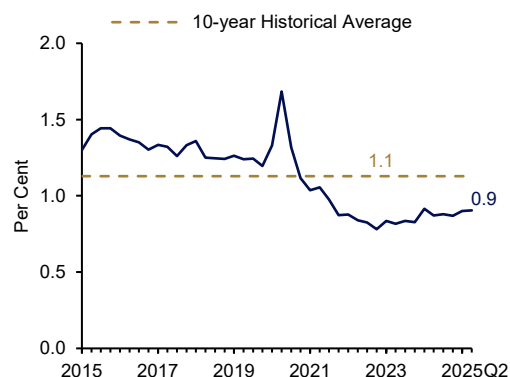
Rollover balances to PDI



Source: DOS, MAS estimates

Chart 4.6 Delinquency rates have crept up, but remained low at less than 1%

Credit/charge card delinquency rate



Source: MAS

4.2 Outlook

Amid the macroeconomic uncertainties, labour demand is expected to soften with nominal wage growth moderating. Against this, MAS' latest stress test of households incorporates a 20% reduction in household income and a 100 bps increase in mortgage rates for loans taken from FIs.¹⁹ The results suggest that a significant majority of households have the capacity to manage an income shock amid higher debt-servicing costs. However, 1% of households could face negative cashflows²⁰ with savings buffers that cover fewer than six months of this shortfall.²¹ Among these vulnerable borrowers, the majority are middle-aged, comprising a mix of lower-income HDB dwellers with tighter cash buffers and upper middle-income segments with sizeable outstanding private housing loans.

To prepare for a more challenging macroeconomic environment, households should therefore exercise prudence in taking up additional leverage and continue to build sufficient liquidity buffers.

¹⁹ The income shock was calibrated to approximate the largest sustained decline observed in low-income households during the early 2000s, while the 100 bps increase would bring mortgage rates to around the elevated levels seen in 2023 and 2024.

²⁰ Negative cashflow refers to a monthly Expenditure-to-Income (ETI) ratio exceeding one, whereby a household's total monthly expenditure—including the servicing of debt obligations—exceeds their monthly income.

²¹ A three-to-six-month savings buffer reflects a common benchmark for financial resilience, viewed as sufficient to cushion temporary income disruptions. See MAS. 2023. "Basic Financial Planning Guide, Monetary Authority of Singapore.", available at: https://www.moneysense.gov.sg/files/Streamlined_Basic_Financial_Planning_Guide_circulate_on_26_Sep_2023_.pdf.

Box D**Update on the Private Residential Property Market**

The pace of private residential property price increases has been more measured this year...

Price increases in the private residential property market have tapered in recent years. The URA Private Residential Property Price Index rose 3.9% in 2024, a moderation from 6.8% and 8.6% in 2023 and 2022, respectively. In 2025, prices have edged up, averaging about 0.9% over the first three quarters (**Chart D1**). This reflects the sustained effects of both demand and supply-side measures. Existing macroprudential and property market measures have restrained excessive demand, while the release of private housing sites via the Government Land Sales (GLS) programme has ensured adequacy of supply of new private housing (**Chart D2**). The private rental market has also stabilised, with rentals remaining broadly flat over the last two years, as the arrival of substantial new supply on the market helped cater to demand.

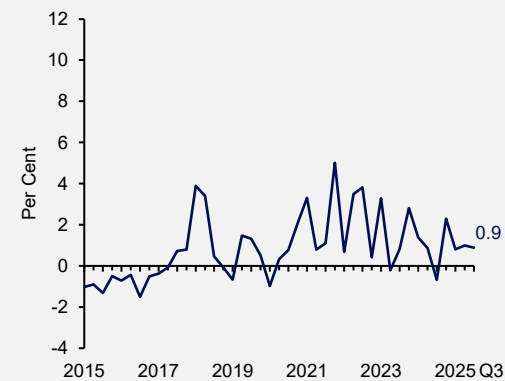
...but transaction volumes have picked up on the back of firmer new sales

Transaction volumes²² picked up recently, driven by a strong recovery in new sales. Average quarterly volumes in the first three quarters of 2025 have exceeded average levels observed over the past three years (**Chart D3**). New sales increased firmly compared to the same period last year as project launches saw firm take-up, while resale volumes also held up.

²² Transaction volume figures exclude Executive Condominiums (ECs), unless indicated otherwise.

Chart D1 Property price growth has remained stable in recent quarters

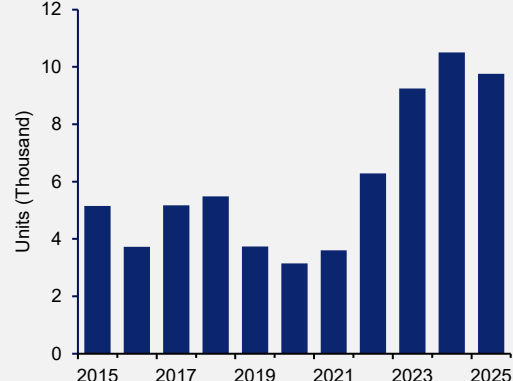
Private residential property price index (q-o-q change)



Source: URA

Chart D2 The government has ensured a steady GLS supply in recent years

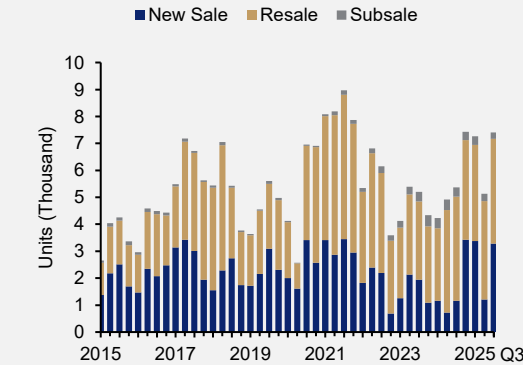
GLS confirmed list supply



Source: URA

Chart D3 Transaction volumes have picked up in 2025

Number of private residential property transactions



Source: URA

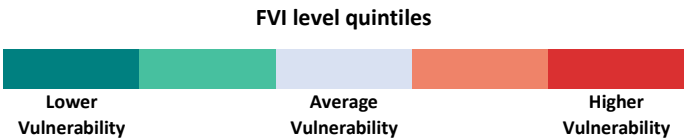
Continued vigilance amid indications of underlying strength in property demand

While there are risks to the macroeconomic environment amid geopolitical tensions and trade policy uncertainties, more accommodative domestic interest rate conditions could support market sentiment and buying interest. The government will remain vigilant to evolving market developments, with the objective of ensuring a stable and sustainable private housing market. Households should continue to exercise prudence in making large financial commitments including property purchases.

5 Singapore Banking Sector

5.1 Assessment of vulnerabilities

Banking sector FVI ²³ level	Q2 2024	Q2 2025
Overall Banking FVI		
Resident leverage risk		
Non-resident leverage risk		
Liquidity risk		
Maturity risk		



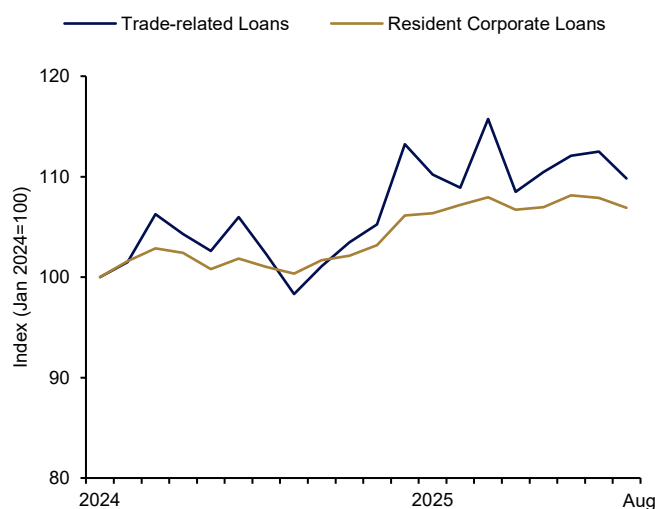
Singapore banks continue to be underpinned by strong capital and liquidity buffers, with the overall banking FVI level remaining benign. Liquidity and maturity risks have stayed low in 2025, with banks maintaining healthy liquidity positions and stable loan-to-deposit (LTD) ratios (see Chart Panel 5B “Banking Sector: Asset Quality and Liquidity Indicators”, Chart 5B4). Leverage vulnerabilities for both residents and non-residents are low amid strong capital buffers.

Credit to non-bank entities expanded at a healthy pace over the past year, supported by strong growth in loans to resident borrowers (see Chart Panel 5A “Banking Sector: Credit Growth Trends”, Chart 5A2). Resident credit growth has been underpinned by lending to the trade-related sectors, which were in turn buoyed by front-loading of exports and strong global demand for electronics linked to AI-related products (**Chart 5.1**).

²³ The banking sector FVI consists of four risk indicators: (i) resident leverage risk assesses Singapore banks’ vulnerability to resident credit growth against banks’ capital buffers that can be used to absorb potential credit losses; (ii) non-resident leverage risk assesses Singapore banks’ vulnerability to non-resident credit growth against banks’ capital buffers that can be used to absorb potential credit losses; (iii) liquidity risk assesses the banking system’s ability to overcome short-term liquidity shocks through indicators such as liquidity coverage ratios; and (iv) maturity risk assesses the banking system’s ability to fund longer-dated assets with short-term liabilities, such as LTD ratios.

Chart 5.1 Resident credit saw robust growth over the past year, supported by loans to trade-related sectors

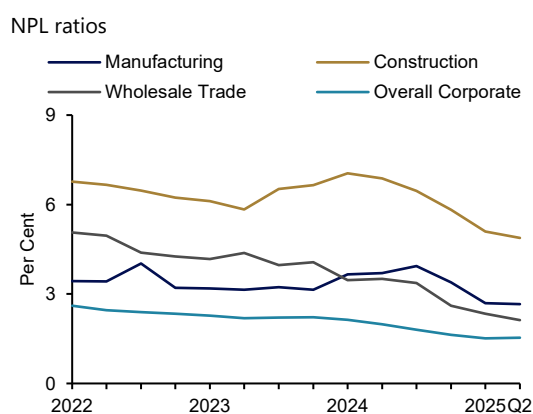
Outstanding loans to resident corporates



Source: MAS

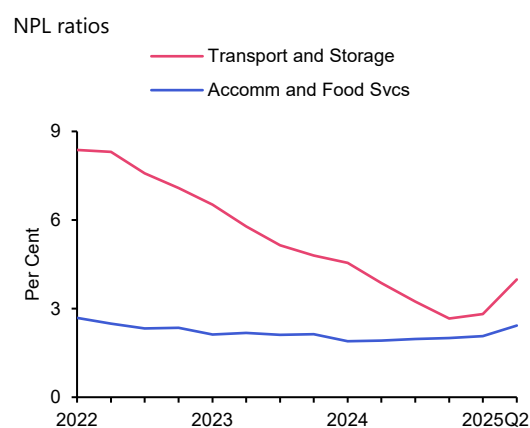
Asset quality has remained generally strong with NPL ratios falling across most sectors (**Chart 5.2**). However, the transport and storage sector, and the accommodations and food services (AFS) sector saw upticks in their NPL ratios, with the latter weighed down by cost pressures and demand shortfalls (**Chart 5.3**).

Chart 5.2 While most sectors saw further declines in NPL ratios...



Source: MAS

Chart 5.3 Transport and Storage and AFS sectors saw a rise in NPL ratios



Source: MAS

Banks continue to maintain adequate provisioning and capital buffers, aided by strong profitability. Banks have strengthened precautionary buffers to guard against a potential rise in credit costs given the uncertain economic environment. The banking system's total

provisioning coverage²⁴ increased to 137% as of Q2 2025 from 113% as of Q2 2024 (see Chart Panel 5B, Chart 5B2).

Local banking groups' robust earnings underpin their strong capital positions, with aggregate Common Equity Tier 1 (CET1) Capital Adequacy Ratios (CAR) well above regulatory requirements (see Chart Panel 5C "Banking Sector: Domestic Systemically Important Banks", Chart 5C1). Strong capital buffers and ample provisions ensure that the local banking groups are well-placed to weather any potential deterioration in credit quality.

5.2 Outlook

An adverse macroeconomic outcome arising from a breakout in trade disputes could impact profitability and cash flows, particularly for firms in trade-related sectors. Although asset quality could come under some pressure going forward, credit risk is expected to be contained with the special mention ratio²⁵ remaining low at 2.5% as of Q2 2025 (see Chart Panel 5B, Chart 5B1).

The results of the IWST 2025 top-down analysis affirm that banks in Singapore have adequate capital buffers to withstand severe macrofinancial shocks (see Box E "Industry-wide Stress Test of Domestic Systemically Important Banks (D-SIBs)"), including further escalation in global trade tensions, a deep and protracted global recession, and the materialisation of a severe global financial stress event.

²⁴ The banking system's total provisioning coverage is the sum of general and specific provisions as a share of unsecured non-performing assets (NPA).

²⁵ Credit facilities that exhibit potential weaknesses but are not yet classified as NPL.

Box E

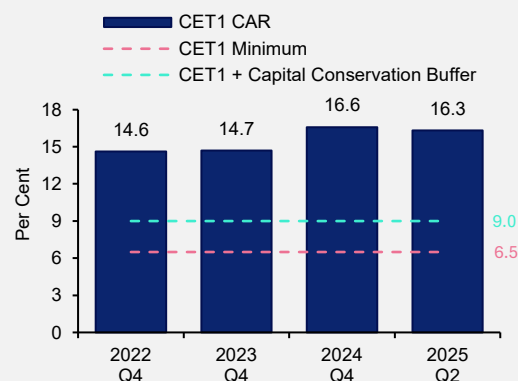
Industry-wide Stress Test of Domestic Systemically Important Banks (D-SIBs)

D-SIBs remain well-capitalised

Singapore's D-SIBs are well-positioned to withstand shocks. This reflects their strong capital buffers, with each D-SIB's CET1 CAR well above the regulatory minimum (**Chart E1**). Moreover, each D-SIB maintains sound credit risk management practices with adequate provisioning buffers to absorb potential credit losses (**Chart E2**).

Chart E1 D-SIB capital positions remain well above regulatory requirements

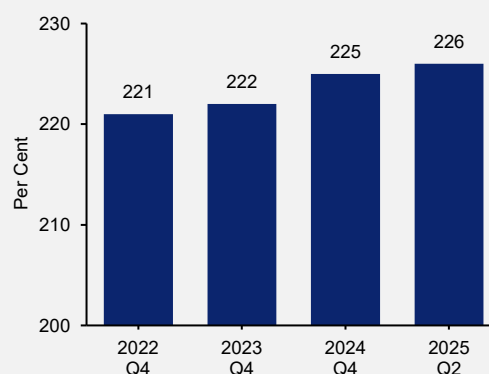
D-SIB aggregate CET1 capital ratios



Source: MAS

Chart E2 D-SIBs maintain sufficient provisions to cushion against credit losses

D-SIB total provisioning coverage ratios[^]



Source: MAS

[^]Total provisions/total unsecured NPA

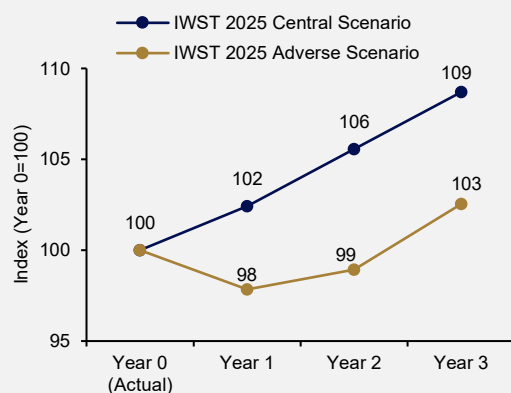
MAS IWST 2025 assessed D-SIB resilience to severe macrofinancial shocks amid an escalation in global trade tensions

The IWST 2025 top-down exercise assessed the resilience of D-SIB balance sheets and capital positions under two scenarios (Central and Adverse) over a three-year horizon. The Central scenario assumes steady global economic growth alongside stable global inflation. Singapore's economy grows at a near-trend pace with inflationary pressures contained.

The Adverse scenario features severe macroeconomic and financial stresses. A sharp retrenchment of global economic activity ensues on the back of escalating trade tensions (**Chart E3**). Singapore slips into a deep and protracted recession (**Chart E4**) amid a severe pullback in external demand, as disruptions to global trade flows and supply chains result in significant production cuts and layoffs by corporates to ensure financial viability.

Chart E3 Under the Adverse scenario of escalating trade tensions, the global economy contracts...

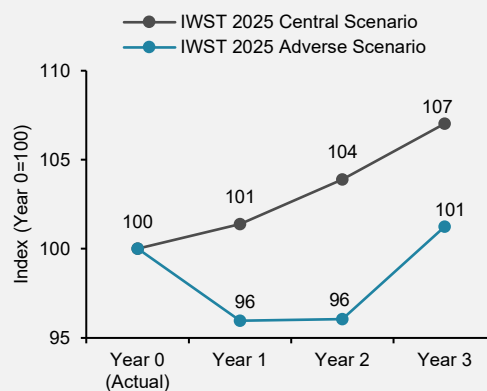
Global real GDP levels



Source: MAS

Chart E4 ... and the Singapore economy enters a deep and protracted recession

Singapore's real GDP levels



Source: MAS

The global recession in the Adverse scenario is compounded by severe and sustained global financial stresses (**Table E1**). Amid escalating trade tensions and elevated fiscal sustainability concerns, market participants reassess asset valuations and reduce their risk appetites. This triggers rapid and broad-based selloffs across financial markets, causing sharp price corrections and disorderly tightening in financial conditions. Elevated credit spreads put pressure on corporate and household debt-servicing capacities, causing banks to incur increased credit losses as defaults rise. As FIs restrict credit supply to preserve capital and liquidity buffers, the sharp pullback in lending to corporates and households further weighs on economic activity.

Table E1 Selected IWST 2025 scenario parameters²⁶

Cumulative change over the IWST scenarios, Year 1 to Year 3		
	Central	Adverse
Equity prices	1% to 14%	-74% to -46%
Property prices	-10% to 18%	-45% to -25%
Central bank policy rates (AE)	-105 bps to -80 bps	-400 bps to -200 bps
Corporate credit spreads	-10 bps to 45 bps	420 bps to 850 bps

Source: MAS

D-SIB aggregate CET1 CAR is projected to remain well above regulatory requirements under the Adverse scenario

Under the Adverse scenario, D-SIB capital positions remain healthy. D-SIB aggregate CET1 CAR is projected to decline over Years 1 and 2, bottoming out at 9.8% in Year 2, but

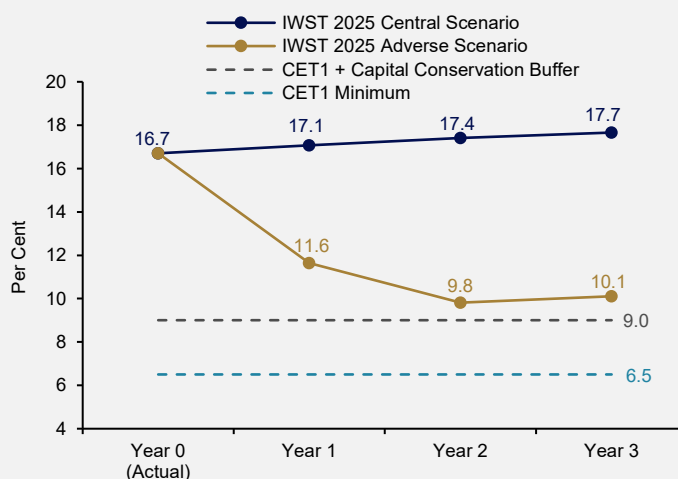
²⁶ Country scope includes G3, ASEAN-5, and EM Asia.

remain above the combined CET1 and capital conservation buffer (CCB) requirement of 9.0% (**Chart E5**). Each D-SIB's CET1 CAR is also projected to remain above this combined regulatory requirement across the scenario horizon.

The decline in the aggregate CET1 CAR is driven by the rise in credit impairments and credit risk-weighted assets (RWA). Against the retrenchment in economic activity and prolonged dislocations in financial markets, the credit quality of corporates and households deteriorate sharply in Year 1. This persists into Year 2, as the Adverse scenario prescribes further increases in credit spreads and sustained declines in equity prices. At the same time, the persistent and significant decline in property prices amplifies the credit risk of loans collateralised by property, translating into higher credit impairments and credit RWAs.

Chart E5 D-SIB aggregate CET1 CAR remains above regulatory minimum under the Adverse scenario

Projected D-SIB aggregate CET1 CAR



Source: MAS estimates

In sum, the IWST 2025 top-down exercise affirmed that the D-SIBs are in a strong position to weather potential shocks even under an Adverse scenario characterised by a deep global recession and protracted severe financial stresses.

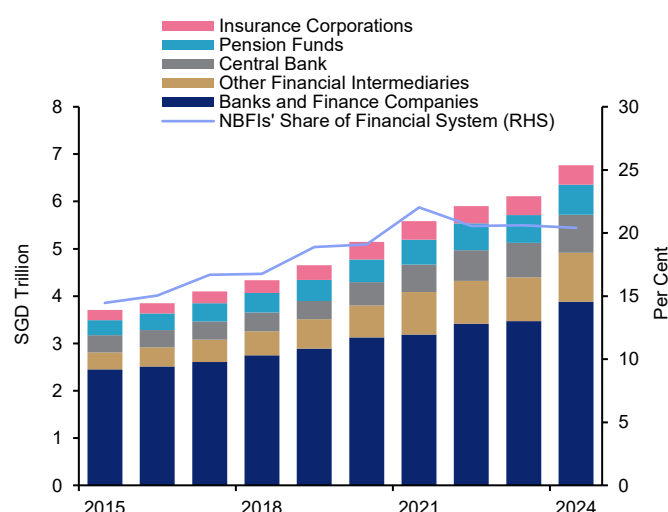
6 Singapore Non-bank Financial Institution (NBFI) Sector

6.1 Assessment of vulnerabilities

The assets of the domestic NBFI sector²⁷ grew firmly by 9.7% over the past year, but its share of total financial system assets declined marginally by 0.2 percentage points to 20.4%, amid the stronger growth in the banking sector (**Chart 6.1**). Most of the expansion in the NBFI sector was in investment funds and trust companies as global equity and credit markets delivered strong returns.

Chart 6.1 Growth in NBFI assets in 2024 was outpaced by banks

Breakdown of financial system assets including NBFIs



Source: MAS estimates

Investment funds

Over the past year, investment funds have managed liquidity risks effectively, meeting redemption requests in an orderly and timely manner despite an increase in such requests after the US announced reciprocal tariffs in April 2025. A large majority of authorised collective investment schemes (CIS)²⁸ maintain cash buffers²⁹ ranging from 0.3% to 11.8%, which is

²⁷ NBFIs refer to Insurance Corporations and Other Financial Intermediaries, which include diverse non-bank entities such as investment funds and trust companies. Prudentially-consolidated brokers are excluded from NBFIs as they are managed as part of banking groups.

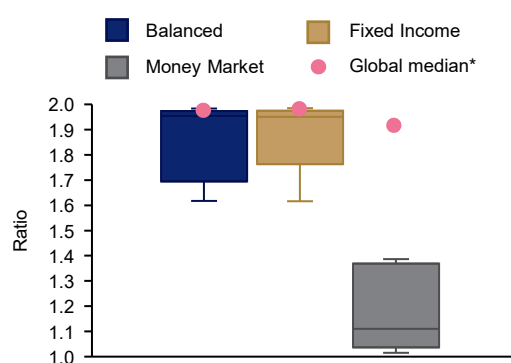
²⁸ Refers to Singapore-constituted collective investment schemes that are offered to retail investors under Section 286 of the Securities and Futures Act 2001 (SFA).

²⁹ Cash holdings (%) is calculated as: cash and deposits / total financial assets

within international industry norms.³⁰ In addition, liquidity mismatches were also low—liquidity transformation ratios (i.e. the degree to which the liquidity of fund assets differs from fund liabilities) for most authorised CIS were below the global median³¹ (**Chart 6.2**).

Chart 6.2 Authorised CIS have managed liquidity risks well...

Vulnerability metrics³² for Authorised CIS: Liquidity transformation



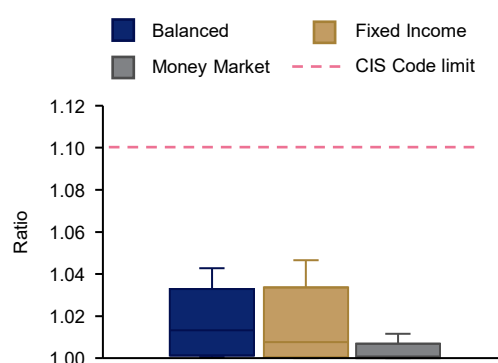
Source: FSB, MAS estimates

Note: Box plots show medians, interquartile ranges and 10th–90th percentiles. Liquidity transformation is calculated as: (Total financial assets – liquid assets [narrow] + short term liabilities [≤30 days] + redeemable equity [≤30 days]) / total financial assets

*Global median is estimated as the median of ratios reported by jurisdictions in the FSB Global Monitoring Report.

Chart 6.3 ... and maintained prudent leverage levels

Vulnerability metrics³³ for Authorised CIS: Leverage



Source: MAS estimates

Note: Box plots show medians, interquartile ranges and 10th–90th percentiles. Balance sheet leverage is calculated as: Total financial assets / equity

Leverage risks have remained broadly manageable for investment funds over the past year. Authorised CIS maintain low leverage ratios averaging 1.01, below the limit of 1.1 imposed by the Code on Collective Investment Schemes (CIS Code)³⁴ (**Chart 6.3**). Most S-REITs also continue to operate within regulatory limits³⁵ with a median leverage ratio of 41%. While hedge funds operating in Singapore saw material increases in the use of leverage as of end-2024,³⁶ the leverage ratios of these funds did not exceed global norms. Supervisory

³⁰ Open-ended funds typically have cash holdings in the range of 0–5% of total net assets. See Schimpf, Andreas, Ilhyock Shim, Hyun Song Shin. 2021. "Liquidity management and asset sales by bond funds in the face of investor redemptions in March 2020." *BIS Bulletin*, No. 39; Yamamoto, Kenta, Tomohiro Okubo, Nobuhiro Abe, Yukio Minoura. 2025. "The Presence of Foreign Open-End Funds in Japan's Financial Markets." Bank of Japan Working Paper, No. 25-E-8.

³¹ See Financial Stability Board (FSB). 2024. "Global Monitoring Report on Non-Bank Financial Intermediation 2024."

³² Vulnerability metrics are designed to assess risks within key NBFI sectors. They are tracked by the FSB and presented in the Global Monitoring Report on NBFI. These metrics are calculated for authorised CIS that were reported via MAS Notice SFA 04-N23 as of June 2025, excluding fund-of-funds.

³³ See footnote 32.

³⁴ Under the CIS Code, authorised CIS (excluding hedge funds and property funds) may borrow only on a temporary basis to meet redemptions and bridging requirements. The borrowing period may not exceed one month. Moreover, aggregate borrowing is capped at 10% of the fund's NAV.

³⁵ Under the CIS code, REITs are subject to an aggregate leverage limit of 50% of its deposited property and a minimum ICR of 1.5 times.

³⁶ Gross leverage of hedge funds rose to approximately 12.5 times of NAV, while financial leverage edged up slightly to about 0.48 times of NAV.

engagements with these hedge funds indicated that they maintained adequate liquidity buffers and practised active risk monitoring throughout stress episodes and were able to meet margin calls and redemption requests.

Insurers

The insurance industry in Singapore remains well-capitalised. The average CARs for direct life and general insurers are well above regulatory requirements (see Chart Panel 6A “Insurance Sector”, Chart 6A1).

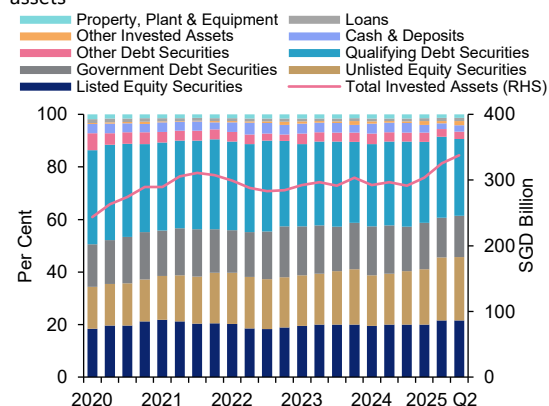
The direct life insurance sector experienced an increase in new business premiums for H1 2025, compared with the same period in the previous year, driven primarily by sales of investment-linked policies (Chart 6A2). Industry profit levels remained stable over the past few quarters (Chart 6A3). The direct general insurance sector also reported solid gross premium growth for H1 2025, driven by business in Singapore (Chart 6A4), coupled with favourable underwriting results and investment income (Chart 6A5). Meanwhile, the reinsurance sector has remained resilient, with steady gross premiums and positive underwriting results (Charts 6A6 and 6A7).

Direct life and composite insurers reported sustained growth in their invested assets, with a slight shift in allocation towards equity securities in recent quarters (**Chart 6.4**). Invested assets of direct general insurers and reinsurers also increased, albeit at a slower pace compared to that of direct life and composite insurers. The asset allocations of direct general insurers and reinsurers have remained relatively stable (**Charts 6.5 and 6.6**).

Credit quality of debt securities held by insurers and reinsurers was good, with over 90% of these securities classified as investment grade. Most of the unrated debt securities were those issued by Singapore statutory boards and Temasek-linked companies.

Chart 6.4 Invested assets of direct life and composite insurers continued to grow

Direct life & composite insurers: Breakdown of invested assets

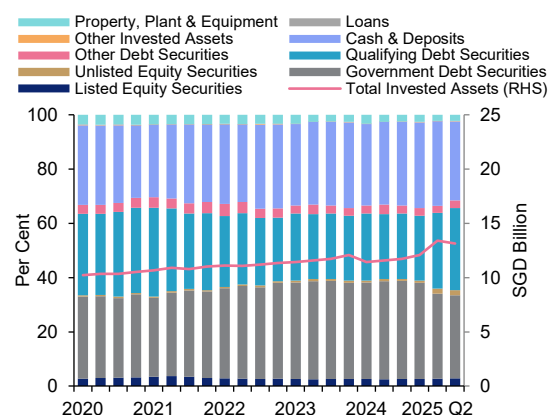


Source: MAS estimates

Note: Equity securities include collective investment schemes and other equity securities, as defined in MAS Notice 129. A qualifying debt security is also defined in MAS Notice 129 and includes debt security that is issued by statutory boards in Singapore or has an investment grade rating.

Chart 6.5 Invested assets of direct general insurers have also increased

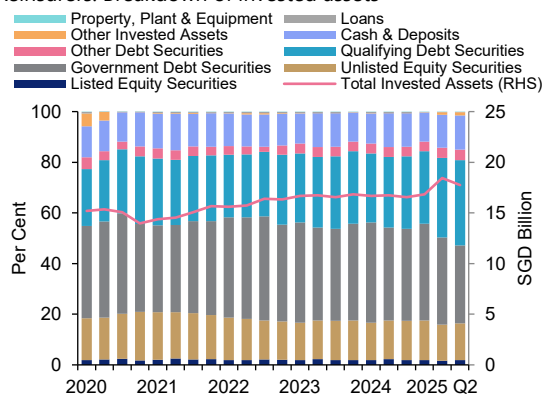
Direct general insurers: Breakdown of invested assets



Source: MAS estimates

Chart 6.6 Invested assets of reinsurers have likewise grown

Reinsurers: Breakdown of invested assets



Source: MAS estimates

Crypto-asset intermediaries and issuers

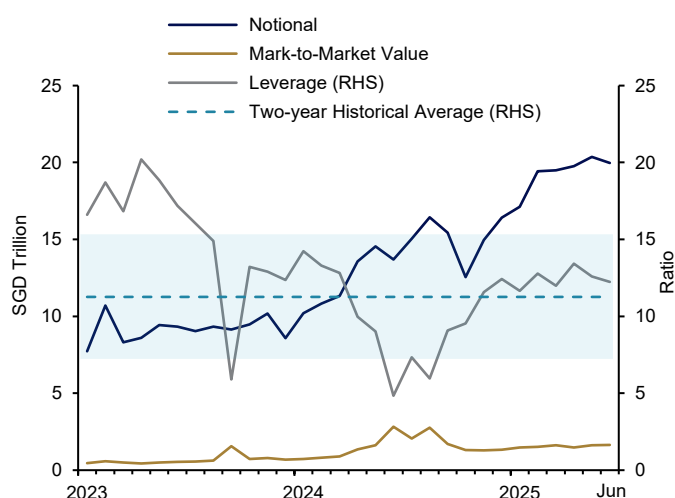
Digital payment token service providers and issuers in Singapore remain relatively small in scale compared to traditional FIs, despite the rapid growth in the digital asset space over the past year. The linkages between crypto-asset players and the broader financial system are limited, and there has been minimal spillover from volatility in the crypto markets into the broader financial system to date. Ongoing surveillance will continue to ensure that growing connections do not introduce risks to financial stability.

Over-the-counter (OTC) Derivatives

Synthetic leverage and gross notional exposures undertaken by NBFIs in the Singapore OTC derivatives markets have increased across most asset classes over the past year, reflecting the global growth in NBFI assets. Synthetic leverage in the OTCD market peaked in April 2025 and moderated in the months after, to reach 12.2 times of gross notional exposure relative to mark-to-market value as of June 2025 and keeping within one standard deviation of its two-year historical average (**Chart 6.7**).³⁷ The decline in synthetic leverage after April 2025 suggests that NBFIs had been deleveraging, likely driven by risk aversion.

Chart 6.7 NBFIs' gross notional exposures and synthetic leverage grew in the first half of 2025

NBFIs' OTC derivatives positions



Source: Depository Trust & Clearing Corporation Data Repository (Singapore) Pte Ltd (DDRS), MAS estimates

Note: Shaded area indicates the one standard deviation range from the two-year historical average of synthetic leverage.

6.2 Outlook

Stress tests of Singapore investment funds and insurers indicate that these sectors are well-positioned to weather global shocks with limited risk spillovers to the rest of the domestic financial system. However, concerns about global macroeconomic uncertainty, elevated asset valuations, and fiscal sustainability present as risk factors for NBFIs in Singapore. These risks may manifest as sharp repricing shocks, with the potential to cause spikes in investor redemptions or impose capital losses on insurers.

Most investment funds have sufficient liquid assets to meet redemptions

The results of MAS' stress simulations³⁸ indicate that most authorised CIS will be able to meet redemptions even under a severe redemption shock, with simulated net outflows

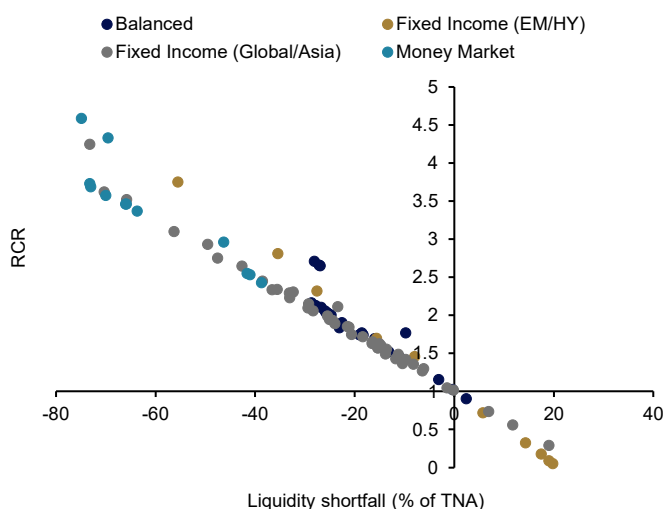
³⁷ Synthetic leverage is proxied by the ratio of the gross notional amount to gross market value of NBFIs' OTC derivatives positions, in line with metrics published by the FSB. See FSB. 2023. "The Financial Stability Implications of Leverage in Non-bank Financial Intermediation".

³⁸ A liquidity stress simulation exercise was conducted for authorised CIS, drawing on enhanced data collection of daily positions and net redemptions of funds as of June 2025. The simulation exercise follows the methodology covered

ranging from 20.8% (EM fixed income funds) to 26.8% (global fixed income funds) of total net assets (TNA). Some fixed income funds and one balanced fund would face liquidity shortfalls, ranging from 2% to 20% of TNA (**Chart 6.8**). However, these funds account for a small proportion (3.8% of TNA) of all the funds assessed.

Chart 6.8 Some fixed income funds may face liquidity shortfalls of up to 20% of TNA under a severe redemption shock

Redemption coverage ratio (RCR) and liquidity shortfall



Source: MAS estimates

Note: The scope of this year's stress simulation was focused on fixed-income authorised CIS given they are more susceptible to liquidity mismatch risks and offered to domestic retail investors.

Insurers can meet regulatory capital requirements under a stress scenario

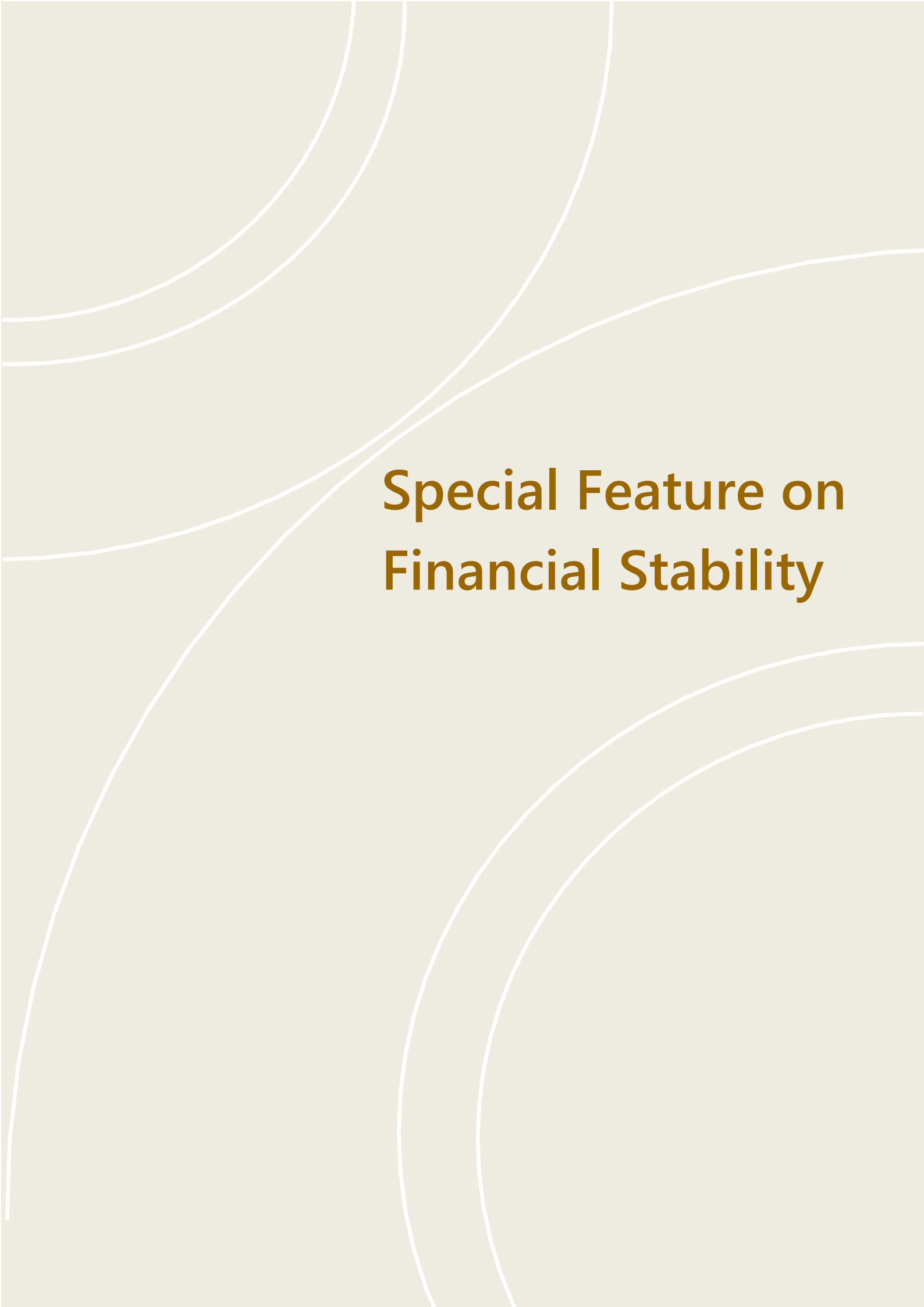
Resilient global economic growth over the past year has supported increases in new business premiums and asset holdings of insurers in Singapore. At the same time, ongoing geopolitical uncertainties could impact global financial markets and insurers' solvency and liquidity positions. Overall, the CARs of insurers in Singapore remain healthy (see Chart Panel 6A, Chart 6A1).

To assess insurers' resilience to shocks, MAS conducted a stress test featuring a Central scenario and an Adverse scenario over a three-year horizon. In the Adverse scenario,³⁹ which assumed declines in interest rates, steep falls in equity prices and credit spread widening, insurers were impacted mainly by the lower valuation of assets arising from the fall in equity prices and credit spread widening, and in the case of life insurers, increased long-term guaranteed liabilities arising from lower interest rates. However, insurers were able to continue

in the 2023 FSR, "Box D: Liquidity Stress Simulation of Investment Funds in Singapore". The redemption shock applied corresponds to the 90th percentile of the distribution of all individual fund-level redemption shocks. Redemption shocks at the individual fund-level were calibrated based on the largest 1% of historical net outflow episodes experienced by the fund.

³⁹ The stress scenario followed scenario parameters in Table E1 (see Box E "Industry-wide Stress Test of Domestic Systemically Important Banks") that were applied to the banks. In addition, insurer-specific parameters were included such as a 100–150% increase in the termination rates for regular premium insurance policies and a 30–100% reduction in the projected new business premiums for participating policies and investment-linked policies.

meeting regulatory capital requirements, in certain cases after undertaking management actions such as reducing bonus allocations, changing asset allocations, reducing dividend transfers to shareholders and seeking capital injections from parent companies.



Special Feature on Financial Stability

Special Feature 1

Exchange Rate Flexibility in Asia and Latin America

Introduction

No matter how committed they may be in principle to a freely floating currency, EM economies often find it difficult to ignore abrupt exchange rate shifts. In responding to external shocks, they might demonstrate a “fear of floating,” the title of a frequently-cited 2002 paper by Guillermo Calvo and Carmen Reinhart. Floating exchange rates offer more monetary flexibility and, frequently, greater capacity to cushion external shocks than fixed rates. But such considerations are likely to be outweighed, in many cases, by concerns about balance sheet exposures and worries about anti-inflationary credibility that motivate central banks to restrict exchange rate flexibility.

Since the 1997–99 Asian financial crisis and the currency crises in Mexico, Brazil, and Argentina during the same era, central banks in both Asia and Latin America have moved in the direction of greater transparency and higher acceptance of floating currencies. Nevertheless, there is a perception that the countries in Latin America have embraced this shift towards greater currency flexibility more fully. Studies such as Ilzetzi, Reinhart, and Rogoff (2019) largely bear this out, classifying four of the five large Latin American inflation targeters as floaters,⁴⁰ while showing Asian countries such as Malaysia, Thailand, and Indonesia as administering de facto crawling pegs.

To gain a better understanding of policy responses to external shocks, the study begins with a brief discussion of reasons for fear of floating, from the perspective of EMs. This is followed by a comparison of descriptive statistics, including average exchange rate variation. A more analytic approach is then introduced, incorporating imperfect asset substitution into the balance-of-payments identity, and using this to break down each country’s response to exogenous shocks. Relative performances of each regime are then assessed, based on variability of inflation and output.

Why fear floating?

Concern about the inflationary impact of large currency movements may motivate a central bank to resist depreciation, particularly if the central bank’s credibility is limited and it is unable to guide inflation expectations. This is often the case if there is a history of currency crises and/or hyperinflation, or a danger of fiscal dominance, as has been prevalent in Latin American and other developing economies. Fear of floating may also be tied to the presence of large dollar obligations on private sector balance sheets. If dollar debts are large enough,

⁴⁰ Brazil, Chile, Colombia, and Mexico. The fifth, Peru, is classified as following a crawling band.

a sudden depreciation can precipitate a wave of bankruptcies and bank failures, threatening financial stability and provoking an economic downturn.

With currency markets often shallow, particularly in smaller economies, currencies may overshoot in either direction in response to disturbances in global financial conditions. An unwarranted depreciation works like a cost-of-living shock, reducing real incomes and imposing financial hardship. But a rapid appreciation may be equally undesirable, bringing on a bout of “Dutch disease” that threatens the competitiveness of export industries and raises unemployment. When economies are highly trade-dependent, as is the case in Asia and elsewhere, the exchange rate is a key reference price, implying that there is a positive benefit in reducing its variability.

These considerations come into play even when the central bank is formally committed to inflation targeting and a floating currency. The case for limiting some exchange rate fluctuations, even under an inflation targeting regime, has strengthened in recent years, as recognised for example in the IMF’s Integrated Policy Framework (Adrian *et al.* 2021). At the same time, these interventions normally entail substantial trade-offs that depend on individual country circumstances, suggesting that the basis for fear of floating is likely to vary from country to country, and may shift over time.

Where have exchange rates varied the most?

There are several ways to assess a central bank’s tolerance for exchange rate fluctuations. Economies are subjected to different exogenous shocks, and the frequency of large shocks varies over time, depending on the business cycle, political conditions, and other factors. Nonetheless, this section begins with some simple objective measures of nominal exchange rate variation, and variation in other monetary variables, including interest rates.

For the seven Asian, five Latin American, and two other emerging economies covered below, Ilzetki, Reinhart, and Rogoff (2019) provide the classifications (see second column of **Table S1.1**). Four of five Latin American countries are classified as either free or managed floaters, but only one of seven Asian currencies (Korea) qualifies as floating. The classifications correspond roughly to the frequency of episodes of at least 2.5% variation in the exchange rate over a one-month period. In general, the countries described as managed (or free) floaters undergo more episodes of substantial nominal exchange rate variation than do countries with a narrow moving band or peg. However, the correlation is not perfect. Notably Mexico is classified as a free floater even though it has a lower frequency of large ($\geq 2.5\%$) exchange rate variation than do the managed floaters Brazil, Chile, and Colombia. Outside Asia and Latin America, Poland is rated as having a moving band, even though it has a higher rate of large fluctuation episodes than Colombia, Chile, and Mexico (this may reflect the złoty’s closer link to the euro).

Table S1.1 Classification of currencies

Region and Country	Classification	Frequency of episodes with $ \Delta_{month} > 2.5\%$	Standard deviation of exchange rate
Latin America			
Brazil	Managed float	56%	8.2%
Mexico	Freely floating	33%	3.6%
Chile	Managed float	45%	3.0%
Colombia	Managed float	39%	3.3%
Peru	Crawling band	8%	5.9%
Asia			
Indonesia	Crawling band	28%	5.5%
Malaysia	Moving band	14%	2.5%
Thailand	Moving band	13%	2.8%
Philippines	Crawling band	14%	2.2%
Vietnam	Crawling peg	2%	1.5%
South Korea	Managed float	26%	3.2%
Hong Kong	Pre-announced peg / currency board	0%	0.2%
Other			
Poland	Moving band	47%	3.5%
South Africa	Freely floating	62%	4.1%

Source: Ilzetzi, Reinhart, and Rogoff (2019), Bloomberg, MAS estimates

Note: Classification of currency follows the Ilzetzi, Reinhart, and Rogoff (2019) methodology. Standard deviation of exchange rates is computed using m-o-m changes.

As a point of comparison, the monthly standard deviations of the exchange rate around the mean appreciation rate are also presented. These do not present any substantial differences with the other metrics, with the possible exceptions of Peru and Indonesia.

While average variation is a useful measure of exchange rate flexibility, it falls short of a complete characterisation of a central bank's monetary/exchange rate policy. A low degree of variation may correspond to a specific policy commitment (such as the Hong Kong currency board), or it may simply reflect a period of market calm and stability.

Classifying exchange rate regimes by the response to external shocks

A deeper measure of the central bank's policy preferences can be had by looking at its reaction to a given shock, and, if possible, comparing that reaction to the actions of other central banks responding to the same or similar shocks.

The central bank's policy options in response to an exogenous shock can be derived from a standard balance of payments equation which is shown in the Annex. It can be expressed as:

$$(\text{Total External Shock}) = (\text{Percent Exchange Rate Depreciation}) + (\text{Interest Rate Increase in Percent}) - (\text{Percent Change in FX Reserves}) \quad (1)$$

This analysis of the central bank's options provides a convenient way of gauging the monetary response to an external shock. Specifically, in response to a negative external shock, the central bank has the option of allowing the exchange rate to absorb the shock in a free float, raising interest rates to defend the currency, or intervening to sell reserves and support the currency. Because the shock is external and the country is small, identification issues are largely avoided.

To get stable estimates of the response magnitudes, a vector autoregression (VAR) in first differences is performed for each country in the study, using monthly data. Accumulated impulse responses are used, to show the full impact over time of the given shock. The global shock variable, CISS, is obtained from the first principal component of a vector of global cyclical variables.⁴¹

A full panel of countries is analysed in this way to compare individual country responses to the same or similar shocks (**Table S1.2**). To maximise comparability, the sample period for each country is set from January 2010 to mid-2025. All data on the exchange rate, short-term interest rate, and reserves sales are monthly. The external shock is scaled so that the average total change on the right-hand side of *Equation 1* is one, but there is variation in the impact of the common shock on each economy.⁴²

Across the sample, the responses are consistent with theory while indicating substantial—and broadly expected—variation by country. Almost all sample countries register a depreciation in response to the negative global shock (one of the two exceptions, Hong Kong, has a currency board and a pegged exchange rate). All but two show reserves sales, and most register a procyclical interest rate increase. A tolerance for floating indicator is constructed for each country by dividing the exchange rate response by the aggregate of the exchange rate, interest rate, and FX reserves sales for that country.

Fear of floating is greater in Asia than in Latin America, based on responses to external shocks. Three of the four countries with the highest tolerance for floating score are in Latin America. Leaving out the Philippines, which registers a muted reaction to the global financial shock, all permit greater flexibility than the highest ranked Asian country, Korea.

⁴¹ For more details, see "Special Feature 1 Asian Capital Flows and the Global Financial Cycle" in the 2024 FSR.

⁴² For example, capital controls might affect the transmissibility, and thus the overall impact, of the external shock. In the limit, if the financial account is completely closed, there may be no impact from the shock.

Table S1.2 Summary statistics of Impulse Response Functions to a shock in CISS

Country	Change in exchange rate	Change in reserves	Change in interest rates	Total change	Floating tolerance
Brazil	0.92**	-0.44**	0.058	1.42	64.7%
Mexico	1.06**	-0.051	0.0026	1.12	95.2%
Chile	0.30**	-0.080	-0.04**	0.34	87.9%
Colombia	1.00**	-0.096	0.039	1.13	88.0%
Peru	-0.012	0.064	0.0059	-0.07	-
Indonesia	0.53**	-0.66**	0.079**	1.27	42.0%
Malaysia	0.46**	-0.42**	-0.00241	0.88	52.7%
Thailand	0.20**	-0.024	0.027*	0.25	79.4%
Philippines	0.20*	0.014	0.028	0.21	93.2%
Vietnam	-0.0018	-0.53	0.059	0.58	-0.31%
South Korea	0.63**	-0.31**	0.00018	0.94	67.3%
Hong Kong	-0.0089	-0.33**	0.020	-0.36	-2.6%
Poland	0.75**	-0.38**	-0.018	1.10	67.6%
South Africa	0.92**	-0.31**	0.043*	1.28	72.2%

Source: Bloomberg, MAS estimates

Note: * and ** indicates statistical significance at 90% and 95% confidence level respectively.

As a more comprehensive exercise, two negative external shocks are combined, the global financial cycle shock and a shock to a representative global interest rate. The response to the joint shock can be characterised as:

$$\begin{aligned}
 &(\text{Total External Shock}) + (\text{Foreign Interest Rate Increase in Percent}) = \\
 &(\text{Percent Exchange Rate Depreciation}) + (\text{Interest Rate Increase in Percent}) - (\text{Percent Change in FX Reserves}) \quad (2)
 \end{aligned}$$

If central banks respond similarly to interest rate and other global financial shocks, the composite of the two shocks can be taken to broaden the range of external shocks used to estimate the central bank's policy response.

The results (**Table S1.3**) are theory-consistent and broadly distributed. For every country in the sample, the exchange rate responds to the composite shock with a depreciation. Similarly, every country conducts net reserves sales, supporting its currency in response to the shock. As with the single global financial cycle shock, three of the four most flexible currencies are in Latin America (the fourth, South Africa, is outside both Asia and Latin America). On the other end of the spectrum, five of the six countries with a floating tolerance index below 60% are located in Asia.

Why the regional disparity in the acceptance of exchange rate flexibility? Both regions have histories of currency crises and attacks on fixed or tightly controlled exchange rates. Yet fear of floating remains significantly stronger in Asia than in Latin America, even after steps in both regions towards greater exchange rate flexibility. One relevant difference is that the Asian economies in the study have significantly higher trade dependence than the Latin American

economies. On average, total trade accounts for 96% of GDP in the Asian economies, while the corresponding figure for the Latin economies is less than half that, at 46%.⁴³ Intraregional trade in Asia is also higher than in Latin America, where much trade is commodity-based and with counterparties outside the region. All else equal, Asian economies face incentives similar to those of a trade union or single currency area such as the EU to minimise exchange rate volatility.

Monetary responses to external shocks are not strong on average, but they still provide insight into the relative ability of individual central banks to conduct countercyclical policy. On average, the response of domestic interest rates to negative shocks is close to zero, but for some economies—notably Indonesia, Vietnam, and Hong Kong—the interest rate response accounts for more than 7% of the overall response to a negative shock. In the case of Hong Kong SAR this may be considered a design feature of the currency board arrangement, which commits the Hong Kong Monetary Authority to sell reserves and contract the domestic money supply in response to outflow pressures, raising interest rates. Consistent with the impossible trinity theorem of international economics, central banks operating with other types of exchange rate pegs or narrow bands are also constrained to some degree from deploying independent monetary policy.

Table S1.3 Summary statistics of Impulse Response Functions to composite external shock

Country	Change in exchange rate	Change in reserves	Change in interest rates	Total change	Floating tolerance
Brazil	0.89**	-0.50**	0.06	1.46	61.3%
Mexico	1.0**	-0.094	0.0052	1.13	91.1%
Chile	0.35*	-0.16	-0.03	0.48	73.6%
Colombia	1.07**	-0.10	0.022	1.20	89.7%
Peru	-0.0037	-0.0058	-0.0031	-0.0010	-
Indonesia	0.57**	-0.71**	0.09**	1.37	41.5%
Malaysia	0.52**	-0.45**	0.0018	0.97	53.3%
Thailand	0.25*	-0.28	-0.000020	0.54	47.1%
Philippines	0.25**	-0.09	0.02	0.37	68.2%
Vietnam	0.0071	-0.62	0.07*	0.70	1.0%
South Korea	0.66**	-0.34**	0.0090	1.00	65.7%
Hong Kong	-0.0062	-0.38**	0.028*	0.40	-1.5%
Poland	0.77**	-0.41**	0.000096	1.19	65.2%
South Africa	0.93**	-0.32**	0.052**	1.30	71.5%

Source: Bloomberg, MAS estimates

Note: * and ** indicates statistical significance at 90% and 95% confidence level respectively. A positive change in exchange rate represents currency depreciation.

⁴³ Calculated by dividing total regional output by total regional trade.

Which is best?

As noted by Jeffrey Frankel (1999), “no single currency regime is right for all countries or at all times”. It is nonetheless possible to compare the performances of economies under different monetary regimes, as determined by the analysis of the previous section. For purposes of general comparison, average growth and inflation rates are presented below (**Table S1.4**), followed by the second moments—the variance of growth and inflation. This evidence is descriptive, and does not justify causal inferences of the form “country X would have done better with a float than a peg”, or vice versa.

Table S1.4 Summary of growth and inflation rates

Region and Country	Average growth rate	Average inflation rate	Std dev (growth rate)	Std dev (inflation rate)
Latin America				
Brazil	1.7	5.9	3.6	3.2
Mexico	2.1	4.4	2.4	2.1
Chile	3.1	4.2	5.0	3.3
Colombia	3.8	5.2	4.8	3.2
Peru	3.7	3.3	3.9	2.5
Asia				
Indonesia	5.2	4.1	1.2	3.2
Malaysia	5.1	2.0	4.8	2.4
Thailand	3.1	1.5	7.3	2.7
Philippines	6.3	3.4	3.0	2.6
Vietnam	6.4	5.0	7.7	5.9
South Korea	3.0	2.1	2.2	2.0
Hong Kong	2.5	2.5	5.8	3.7
Other				
Poland	3.9	3.8	3.6	5.1
South Africa	1.6	5.2	2.7	1.9

Source: Haver Analytics, MAS estimates

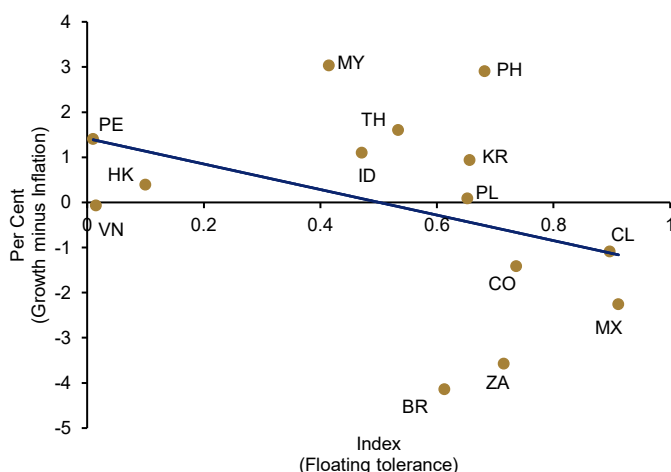
Note: Growth and inflation rates were computed based on annualised q-o-q changes.

Average growth and inflation rates are generally better among the non-floaters (**Chart S1.1**) and demonstrated by a simple regression of the growth-inflation differential⁴⁴ on the floating tolerance measure. This outcome can be attributed to the predominance of Latin American countries among the floaters, as described in the preceding section, and the poorer average growth and inflation performance of countries in that region. Various factors might help explain the difference, including the larger share of commodities in Latin exports, the generally greater trade-dependence of Asian economies, and the weaker fiscal positions of countries in Latin America. Most large Latin American economies, including Brazil, Colombia, and Chile, previously had less flexible monetary and exchange rate policies and greater fear

⁴⁴ Average growth minus average inflation. This is presented as a “satisfaction index” or the inverse of the “misery index.”

of floating, but opted to transition to inflation targeting regimes as their central banks gained credibility. This revealed preference suggests these countries do not view a policy of inflation targeting with a flexible exchange rate as entailing a major performance penalty.

Chart S1.1 Linear regression of growth-inflation differential on floating tolerance index



Source: Bloomberg, Haver Analytics, MAS estimates

Note: Growth and inflation rates were computed based on annualised q-o-q changes.

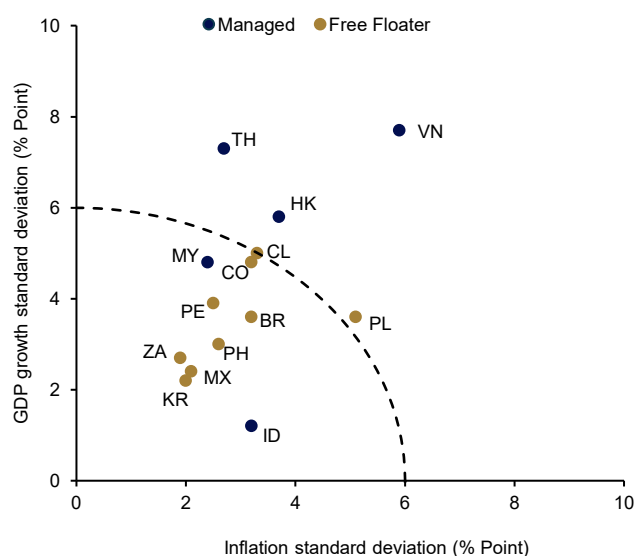
Monetary policy is, in any case, focused more on the business cycle than on long-term growth, which is predominantly a function of structural and fiscal conditions. And while the central bank can clearly influence inflation over both the short- and long-term, its average inflation target is typically decided outside the bank, e.g., by a country's legislature. Accordingly, central banks are seen as being more concerned with second than with first moments—that is, with the divergence of both inflation and growth from targets or long-term values.

Following Svensson (2009), this joint divergence is evaluated by taking the sum of sample variance in output growth and sample variance in inflation, $V = \sum \left(\frac{\dot{Y}}{Y} - \bar{\frac{\dot{Y}}{Y}} \right)^2 + \sum (\pi - \bar{\pi})^2$, which the central bank would like to minimise. Charting the standard deviation of inflation on the x-axis and the standard deviation of growth on the y-axis, a series of concentric circles centred on the origin describes points of equal value of V (**Chart S1.2**).⁴⁵

Evaluated on this criterion, the advantage shifts to the relative floaters. Adopting as a reference the quarter-circle intersecting a standard deviation of six for both growth and inflation, most of the economies closer to the origin, and thus with lower values of V , rank more highly (above 60%) on the floating tolerance scale.⁴⁶ The relative success of the floaters here can be attributed to lower variance of GDP, which is consistent with the idea that a floating exchange rate is more effective than a peg in cushioning external shocks.

⁴⁵ These correspond to the formula for a circle centred at the origin of the Cartesian plane, $x^2 + y^2 = k$

⁴⁶ The two exceptions, Indonesia and Malaysia, have floating tolerance scores of 47% and 42%, respectively.

Chart S1.2 Standard Deviations of Growth and Inflation

Source: Bloomberg, Haver Analytics, MAS estimates

Note: Growth and inflation rates were computed based on annualised q-o-q changes. Countries with a floating tolerance score above 60% are classified as free floaters.

Conclusion

Analysis of the impact of external shocks on EMs shows that the *de facto* policies and preferences of the central bank can vary somewhat from official or published policies. They can also change over time, not necessarily due to statutory changes but rather to adjustments in operational frameworks that make policies more effective.

While generally favourable to floating exchange rates and inflation targeting, the results presented here are a useful reminder that central banks in EM countries do not in practice remain indifferent to exchange rate fluctuations. Although central banks in Latin America have more tolerance for floating than Asian central banks on average, all sell FX reserves in response to negative external shocks, at least at times.

Countries sometimes progress from exchange rate pegs to inflation targeting, which normally requires strengthening macro frameworks and establishing the anti-inflationary credibility of the central bank. Progress in this area, and the lower variability of inflation and growth under inflation targeting, may help to explain the overall shift towards more flexible regimes.⁴⁷ This broader approach, centred on an inflation target as a nominal anchor while conducting occasional interventions to address market imperfections, recalls the principles of the IMF's Integrated Policy Framework.

Different regimes may be more appropriate, or even more conducive to growth, in different circumstances. Joining a trade area or trade union could reduce incentives for floating. Changing global conditions could also affect the calibration of monetary and exchange rate policy. If, for example, the policy framework in the country supplying the global

⁴⁷ See Chapter 2 of IMF World Economic Outlook, October 2025.

anchor currency were to weaken, incentives would shift towards inflation targeting or some other anchor.

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Annex – Derivation of Central Bank Policy Response Equations

The central bank's policy options in response to an exogenous shock can be derived from a standard balance of payments equation:

$$\Delta FR = CA(e, Y) + FA(i, e, i^*, E(e), \varphi) \quad (1)$$

where ΔFR represents the change in FX reserves, $CA(.,.)$ is the current account and $FA(.,.,.)$ is the financial account (BOPM 5 standard). In this simplified representation the current account is a linear function of the (log of the) exchange rate e and output Y :

$$CA = -\beta_1 Y + \beta_2 e$$

where β_1 and β_2 are positive, and e is the exchange rate in units of local currency per dollar (so a rise in e corresponds to a local currency depreciation). The financial account is determined by a modified uncovered interest parity condition and an external shock variable φ :

$$FA = \gamma(i - i^* + e_t - E(e_{t+1}) - \varphi)$$

where i is the nominal interest rate in percentage points, and e_t and e_{t+1} are natural logs of the exchange rate. Expression (1) can be rearranged with the shock variable φ on the LHS and written as:

$$-\varphi = \frac{1}{\gamma}(\Delta FR) - \frac{\gamma + \beta_2}{\gamma}e_t - i + i^* + E(e_{t+1}) - \beta_1 Y$$

Assuming that i^* , Y , and $E(e_{t+1})$ are fixed over the short term, that γ is equal to the stock of FX reserves, and that β_2 is small relative to γ , differencing of the equation yields:⁴⁸

$$d\varphi = -d\left(\frac{\Delta FR}{FR}\right) + de_t + di$$

As sensitivity to different types of external shocks may vary by country, a second version of the exercise was conducted using a proxy for shocks to the foreign interest rate i^* . The external shock variable in this case is the 5-year forward US overnight interest rate. It is selected because changes in a long-term forward rate from one month to the next are likely to reflect new information received during the month, rather than a previously expected shift in short-term rates. Holding Φ constant this time, this yields the following variation on the preceding equation:

$$di^* = -d\left(\frac{\Delta FR}{FR}\right) + de_t + di$$

The results (not shown) from this exercise varied somewhat from the VARs conducted with the global financial cycle variable, but the magnitude and direction of response was largely in accord with priors and theory.

⁴⁸ The $d(\Delta FR/FR)$ term can be interpreted simply as $(\Delta FR/FR)$ if the change in reserves in $(t-1)$ is zero.

As a more comprehensive exercise, the two negative external shocks are combined. Assuming as above that Y and $E(e_{t+1})$ are fixed over the short term, the specification for the response function becomes:

$$di^* + d\varphi = -d\left(\frac{\Delta FR}{FR}\right) + de_t + di$$

As described in the footnote,⁴⁹ Φ is scaled so that, on average, a one-unit shock to Φ changes the total of RHS variables by one. By derivation, a one-unit shock to i^* should change the sum of the same RHS variables by one. Assuming that central banks respond similarly to interest rate and other global financial shocks, the composite of the two shocks can be taken to broaden the range of external shocks used to estimate the central bank's policy response.

⁴⁹ To maintain consistency with the derivation, and to permit comparability with a subsequent shock to the foreign interest rate i^* , the shock variable Φ should be scaled so that a one-unit change in Φ corresponds to a one-unit change in the sum of the RHS variables. However, since there is variation in the degree of individual country responses to the common shock, this is done by taking the initial cross-country average (ICCA) of the sum of RHS variables in response to the common shock and using this to scale the initially arbitrarily indexed financial cycle variable Φ' so that $\Phi = \Phi' \cdot \text{ICCA}$.

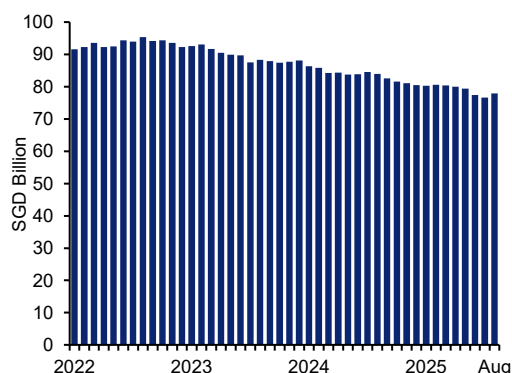


Additional Charts

Chart Panel 3A Small and Medium-sized Enterprise Financing Conditions

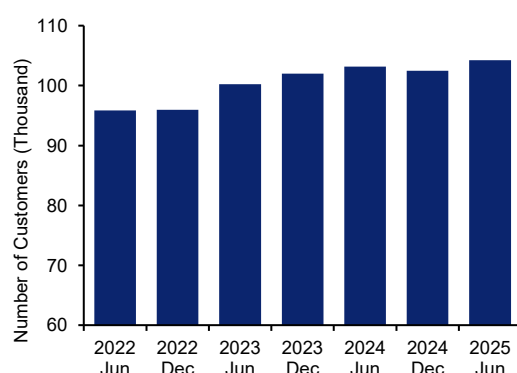
Credit to SMEs continued to ease, reflecting the natural runoff of loans under enhanced ESG loan schemes introduced during the pandemic. The number of SME customers increased.

Chart 3A1 Outstanding SME loans



Source: MAS

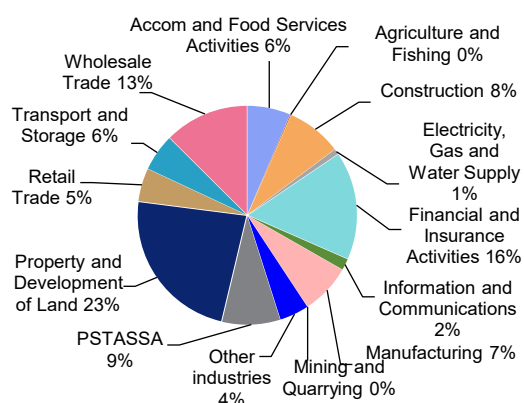
Chart 3A2 SME customers



Source: MAS survey

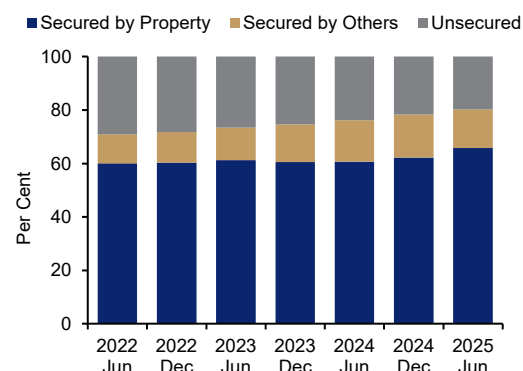
The Property and Development of Land sector continued to account for the largest share of non-financial corporate SME loans. An increasing share of SME loans was collateralised by property.

Chart 3A3 SME loans by sector (as of August 2025)



Source: MAS

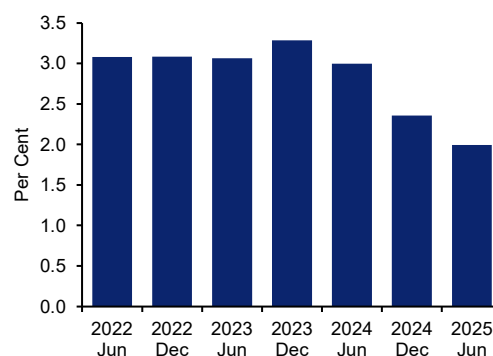
Chart 3A4 Outstanding SME loans by type of collateralisation



Source: MAS survey

SME NPL ratio remained low as of Q2 2025.

Chart 3A5 SME NPL ratio

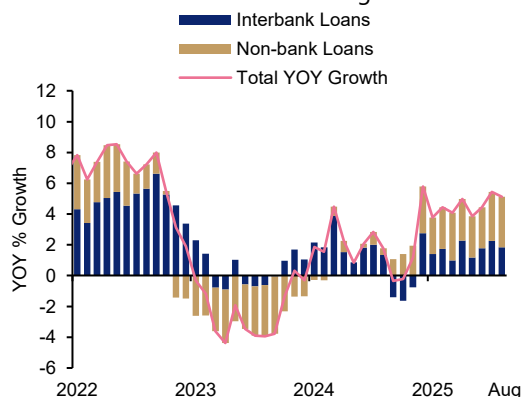


Source: MAS survey

Chart Panel 5A Banking Sector: Credit Growth Trends

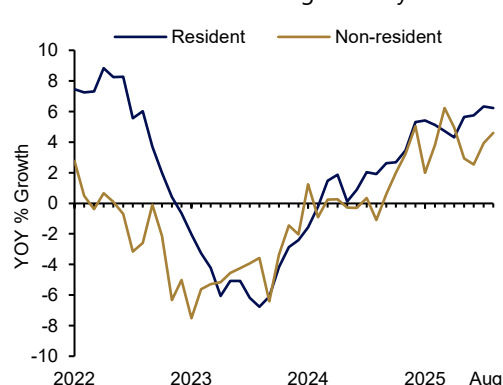
Credit to non-bank entities expanded over the past year, supported mainly by loan growth to residents.

Chart 5A1 Overall loan growth



Source: MAS

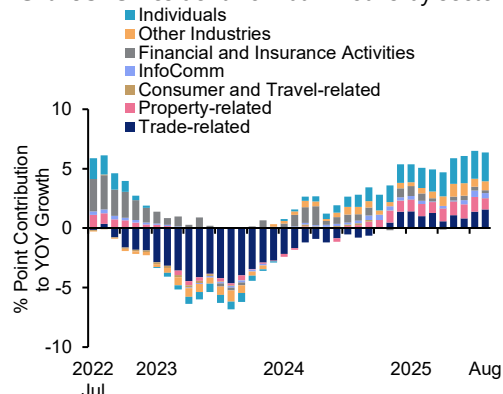
Chart 5A2 Non-bank loan growth by residency



Source: MAS

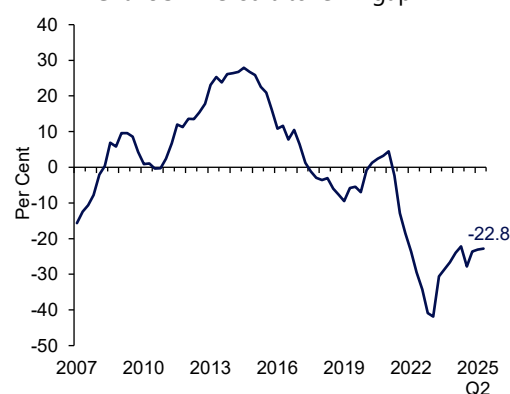
Growth in resident non-bank credit was broad-based, with increased contribution from trade-related sectors in recent months. The credit-to-GDP gap for Singapore remained negative in Q2 2025.

Chart 5A3 Resident non-bank loans by sector⁵⁰



Source: MAS

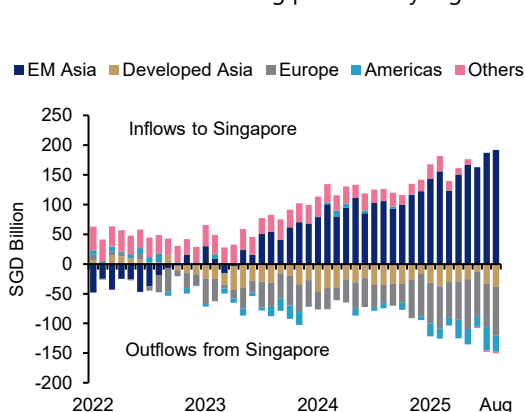
Chart 5A4 Credit-to-GDP gap⁵¹



Source: MAS

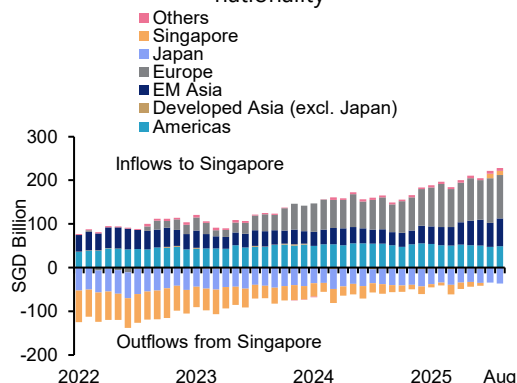
Banks in Singapore continued to intermediate funds from Emerging Asia to Developed Asia and Europe. European banks were the largest intermediators of funds from Emerging Asia.

Chart 5A5 Net lending positions by region



Source: MAS

Chart 5A6 Net lending positions to EM Asia by bank nationality



Source: MAS

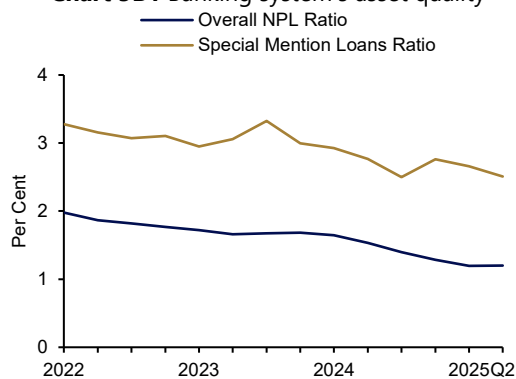
⁵⁰ Consumer and Travel-related refers to Retail Trade and Accommodation and Food Services Activities. Property-related refers to Construction and Property and Development of Land. Trade-related refers to Manufacturing, Wholesale Trade, and Transport and Storage.

⁵¹ The credit-to-GDP gap is defined as the difference between the credit-to-GDP ratio and its long-term trend.

Chart Panel 5B Banking Sector: Asset Quality and Liquidity Indicators

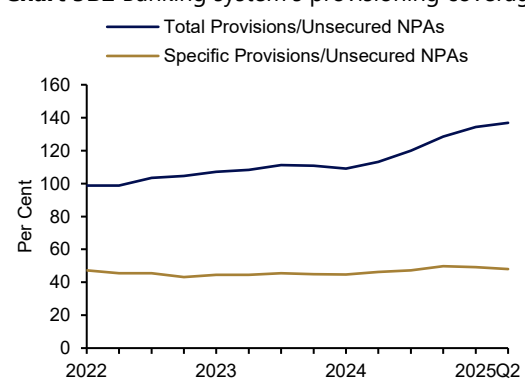
The overall NPL ratio declined to a low of 1.2% as of Q2 2025. Total provisioning coverage increased further to 136.8% as of Q2 2025.

Chart 5B1 Banking system's asset quality



Source: MAS

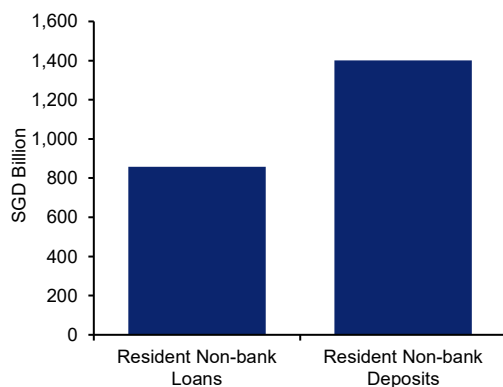
Chart 5B2 Banking system's provisioning coverage



Source: MAS

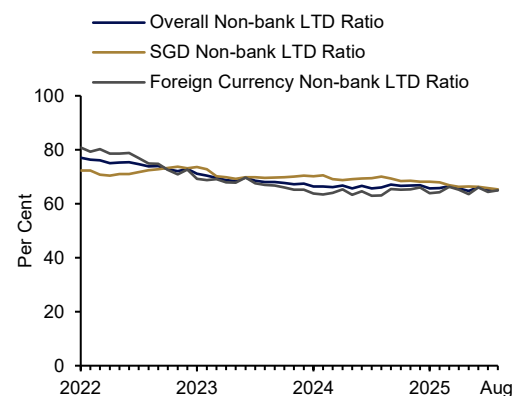
Banks' liquidity positions are healthy. Resident deposits are more than sufficient to fund resident loans. All LTD ratios remained stable and below 100% as of August 2025.

Chart 5B3 Banking system's resident non-bank loans and deposits (as of August 2025)



Source: MAS

Chart 5B4 Banking system's non-bank LTD ratios

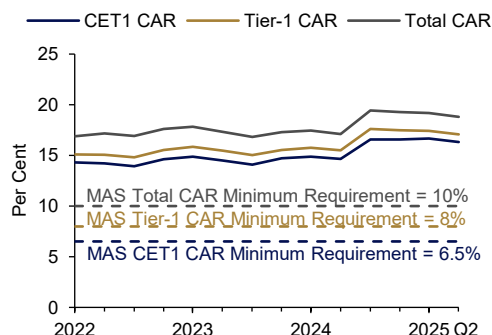


Source: MAS

Chart Panel 5C Banking Sector: Domestic Systemically Important Banks

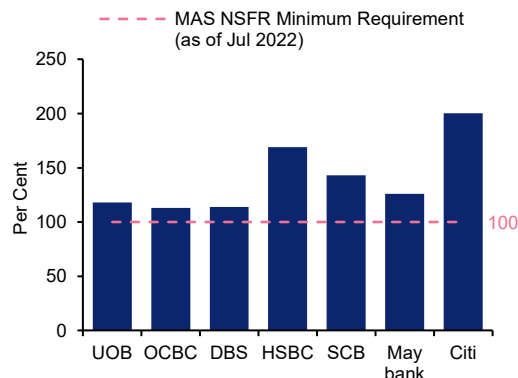
D-SIBs maintained strong capital positions, with aggregate CARs well above regulatory requirements as of Q2 2025. D-SIBs' amount of stable funding was more than sufficient to meet stable funding needs.

Chart 5C1 D-SIBs' CARs



Source: D-SIBs' financial statements, MAS

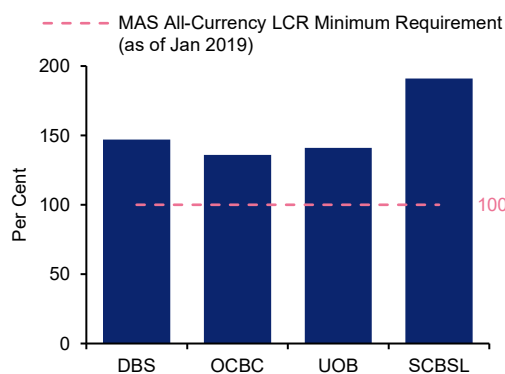
Chart 5C2 D-SIBs' NSFRs (as of Q2 2025)



Source: D-SIBs' financial statements, MAS

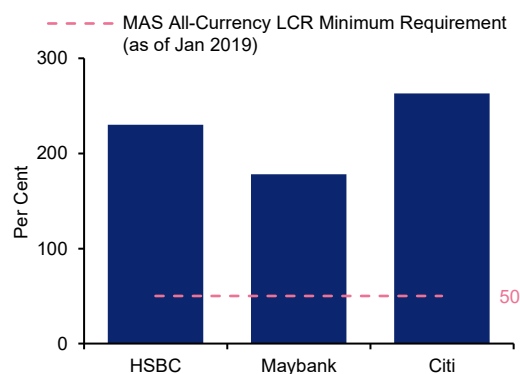
D-SIBs maintained healthy liquidity positions that were well above regulatory requirements.

Chart 5C3 Local banking groups⁵² and SCBSL all-currency LCRs (Q2 2025)



Source: Local banking groups' and Standard Chartered Bank (Singapore) Limited's financial statements, MAS

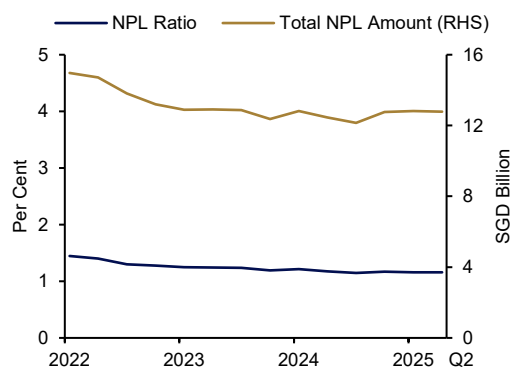
Chart 5C4 Other D-SIBs' all-currency LCRs (Q2 2025)



Source: Other D-SIBs' financial statements, MAS

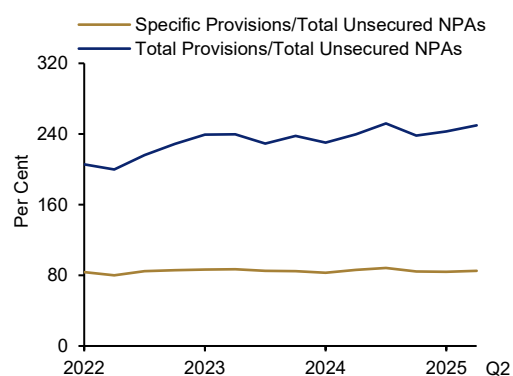
Local banking groups' asset quality remained strong, with NPL ratio at 1.2% as of Q2 2025. Local banking groups maintained a healthy total provisioning coverage of 249.8%.

Chart 5C5 Local banking groups' NPLs



Source: Local banking groups' financial statements, MAS

Chart 5C6 Local banking groups' provisioning coverage



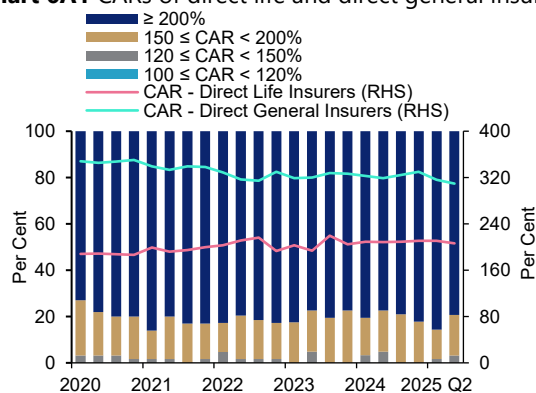
Source: Local banking groups' financial statements, MAS

⁵² Local Banking Groups refer to DBS Bank, Overseas Chinese Banking Corporation and United Overseas Bank.

Chart Panel 6A Insurance Sector

The insurance industry in Singapore remains well-capitalised. Average CARs for direct life and general insurers are well above regulatory requirements.

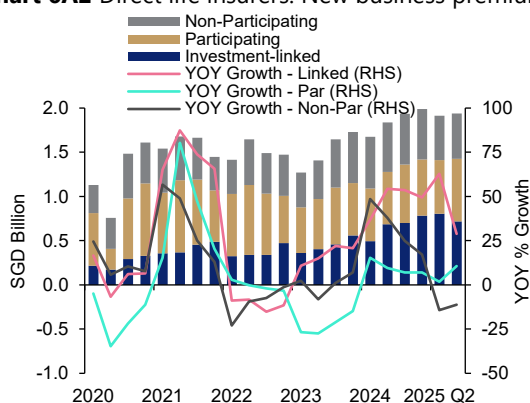
Chart 6A1 CARs of direct life and direct general insurers



Source: MAS estimates

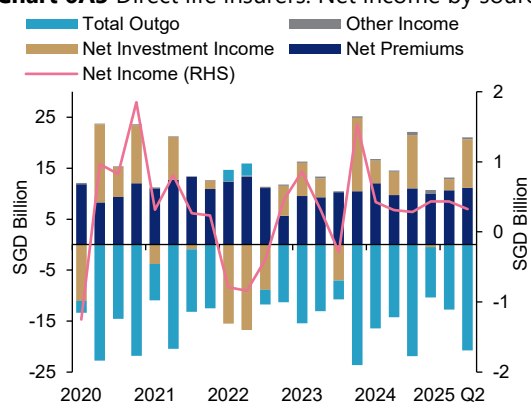
Direct life insurers continue to report robust new business premiums and maintained stable profit levels.

Chart 6A2 Direct life insurers: New business premiums



Source: MAS estimates

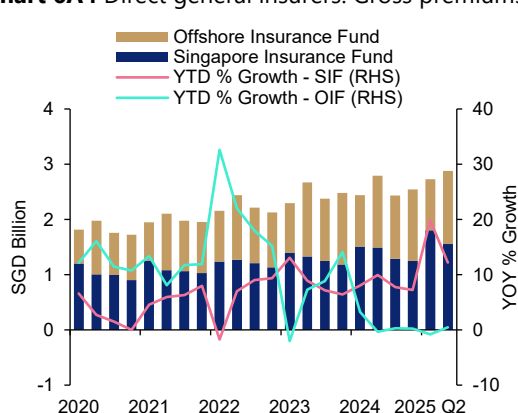
Chart 6A3 Direct life insurers: Net income by source



Source: MAS estimates

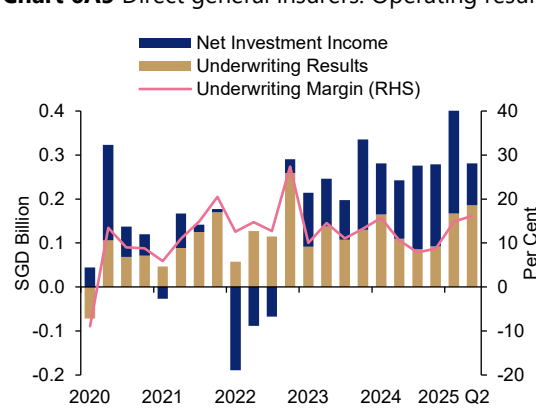
Direct general insurers continue to report solid premium growth, coupled with positive underwriting results and investment income.

Chart 6A4 Direct general insurers: Gross premiums



Source: MAS estimates

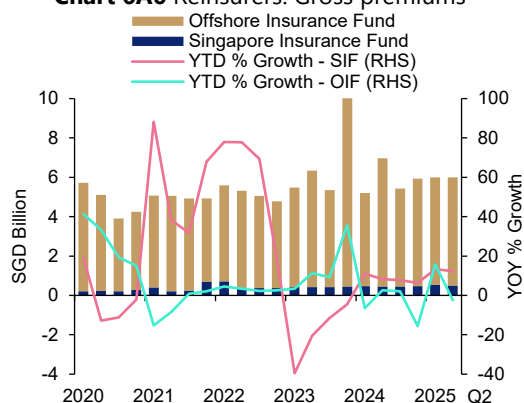
Chart 6A5 Direct general insurers: Operating results



Source: MAS estimates

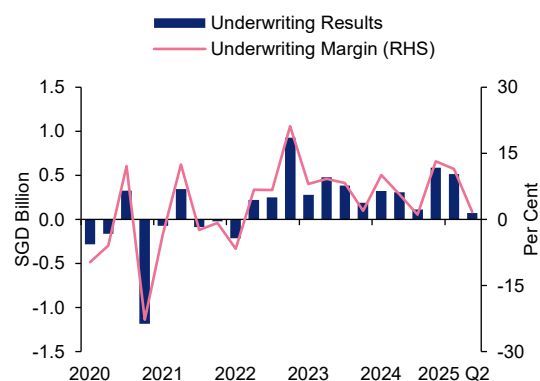
Reinsurers continue to report steady gross premiums and positive underwriting results.

Chart 6A6 Reinsurers: Gross premiums



Source: MAS estimates

Chart 6A7 Reinsurers: Operating results



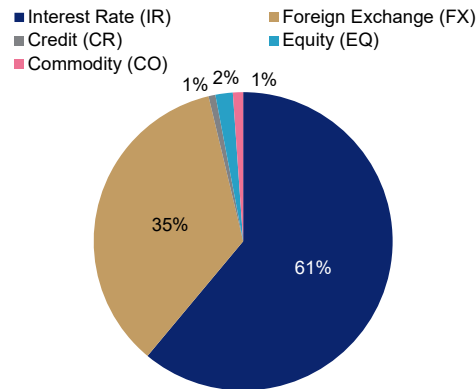
Source: MAS estimates

Note: Underwriting results reflect the profitability of reinsurers' core operations, taking into account the underwriting results reported for general reinsurance business and technical results reported for life reinsurance business, as defined in MAS Notice 129.

Chart Panel 6B Over-the-Counter (OTC) Derivatives

IR and FX continued to be the largest asset classes, based on outstanding notional amounts. Foreign banks or dealers accounted for most trades, either conducted with each other or with NBFIs.

Chart 6B1 OTC derivatives market in Singapore by asset class and notional outstanding (end-Jun 2025)



Source: DDRS, MAS estimates

Chart 6B2 Cross-sectoral breakdown of notional outstanding (end-Jun 2025) – All asset classes

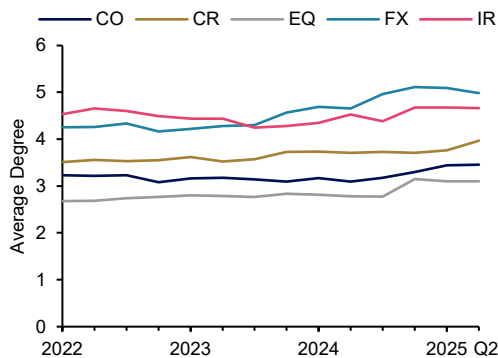
%	LCL BK	FGN B/D	NBFI	C/O	CCP	INDIV
LCL BK	0.3	3.1	0.4	0.2	2.1	0.3
FGN B/D		29.3	28.8	1.9	28.8	0.5
NBFI			0.4	0.2	3.1	<0.1
C/O				0.4	<0.1	

Source: DDRS, MAS estimates

Note: LCL BK = Locally incorporated bank; FGN B/D = Foreign bank or dealer; C/O = corporate or other non-bank entity

Most asset classes have become slightly more interconnected, while some asset classes, especially IR, have become more concentrated as well.

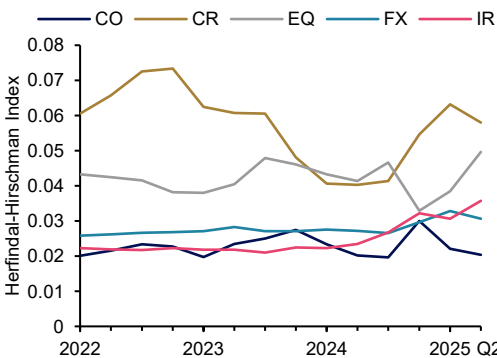
Chart 6B3 Average degree



Source: DDRS, MAS estimates

Note: Degree refers to the number of connections a counterparty has. Connections are counted based on outstanding positions.

Chart 6B4 Herfindahl-Hirschman Index (HHI)

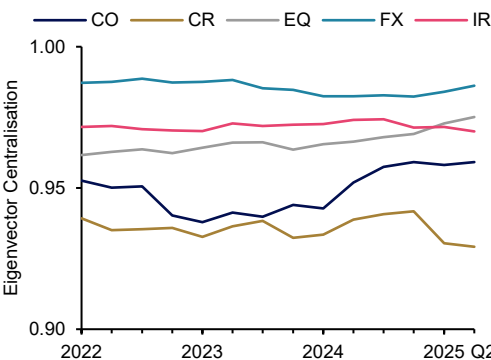


Source: DDRS, MAS estimates

Note: HHI is calculated based on the sum of the squares of the proportion of notional outstanding of each counterparty. Intragroup trades and trades with CCPs have been excluded.

The OTC derivatives network in Singapore is centralised among relatively few influential counterparties, as reflected by high centralisation values.

Chart 6B5 Eigenvector centralisation



Source: DDRS, MAS estimates